# Inżynieria układów i systemów scalonych - projekt

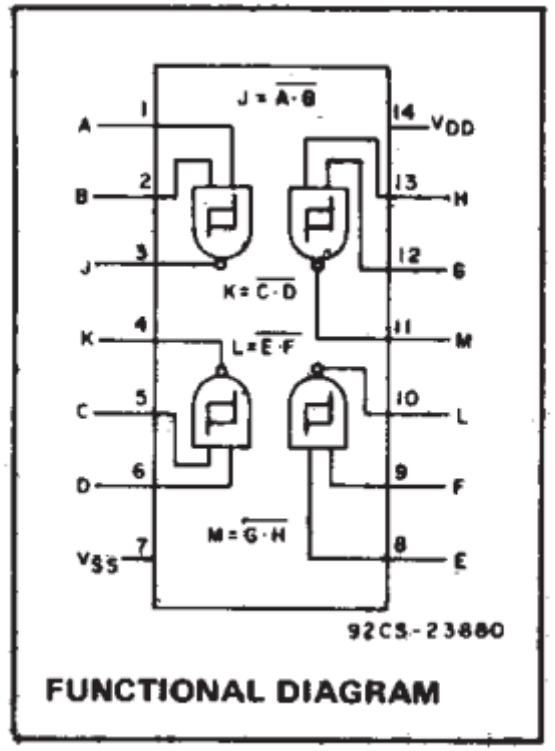
Bartosz Chełmiński 198166

# Lista literatury


- [1] B. Pankiewicz, materiały wykładowe „Inżynieria układów i systemów scalonych”, Gdańsk 2009
- [2] Texas Instruments, nota katalogowa układu CD4093B.
- [3] J. M. Rabaey, B. Nikolic, A. P. Chandrakasan — Digital Integrated Circuits — A Design Perspective (2nd Edition)
- [4] M. Thiele, J. Lienig — Fundamentals of Electromigration-Aware Integrated Circuit Design

# Realizowany układ: CD4093B

4 bramki NAND z histerezą



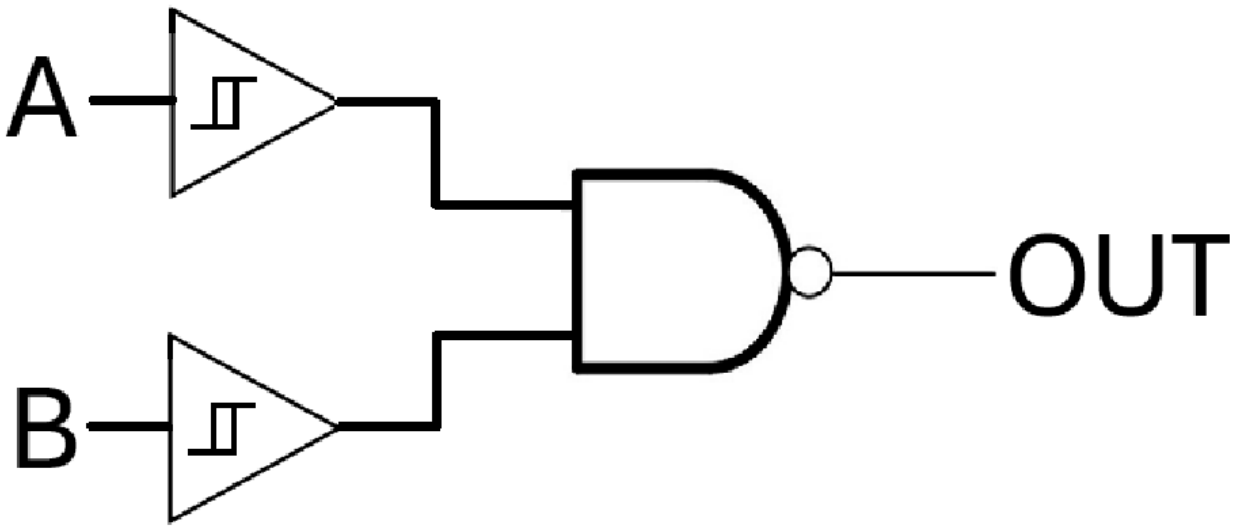

FUNCTIONAL DIAGRAM

# Wymagania projektowe

1) Technologia: CMOS TSMC 180 nm 1P6M 1.8V
2) Zabezpieczenie przed zatrzaskiwaniem
3) Zabezpieczenie przed elektromigracją
4) Pola PAD o wymiarach (szerokość × długość) 60 μm × 70 μm w wersji no-devices-under-pad
5) Zabezpieczenie ESD w postaci diod lub w postaci tranzystorów
6) Wydajność prądowa wyjść u.s.: IO ≥ 15 mA DC w warunkach VDD = 1,8 V +/- 10% i w temperaturze od −30 st. C do +125 st. C
7) Częstotliwość na wyjściach u.s. co najmniej 30 MHz przy obciążeniu 25 pF, zasilaniu 1,8 V +/- 10% i w temperaturze od −30 st. C do +125 st. C.

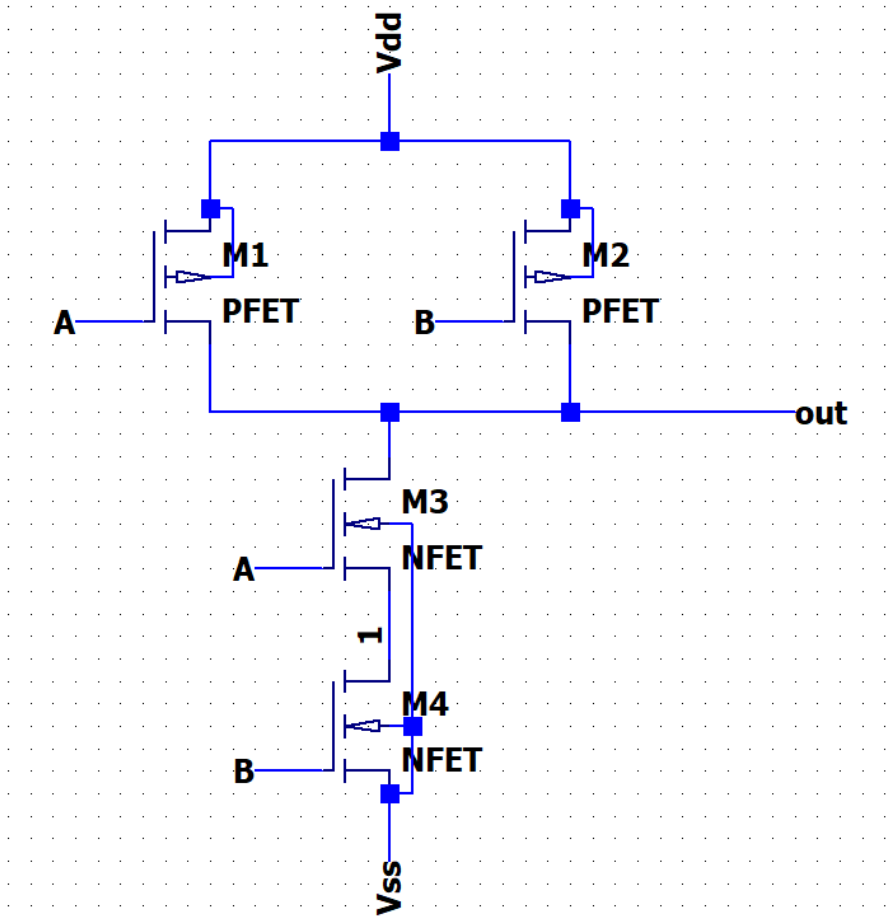
# Zaproponowany schemat pojedynczej bramki

Schemat blokowy

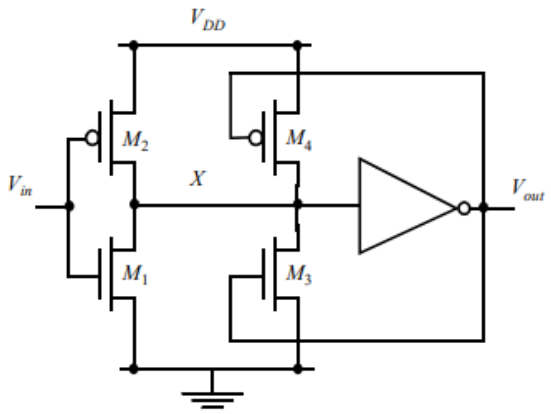


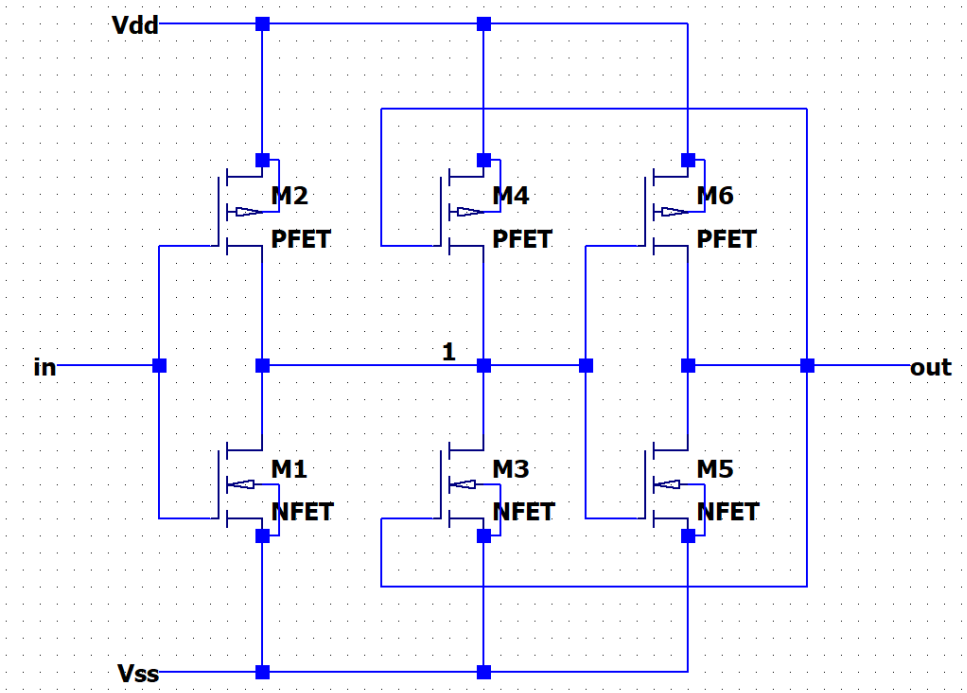
# Bramka NAND

```
.subckt NAND A B out vdd vss
M1 vdd A out vdd PFET l=.2u w=.8u
M3 out A 1 0 NFET l=.2u w=.4u
M2 vdd B out vdd PFET l=.2u w=.8u
M4 1 B 0 0 NFET l=.2u w=.4u
.ends
```



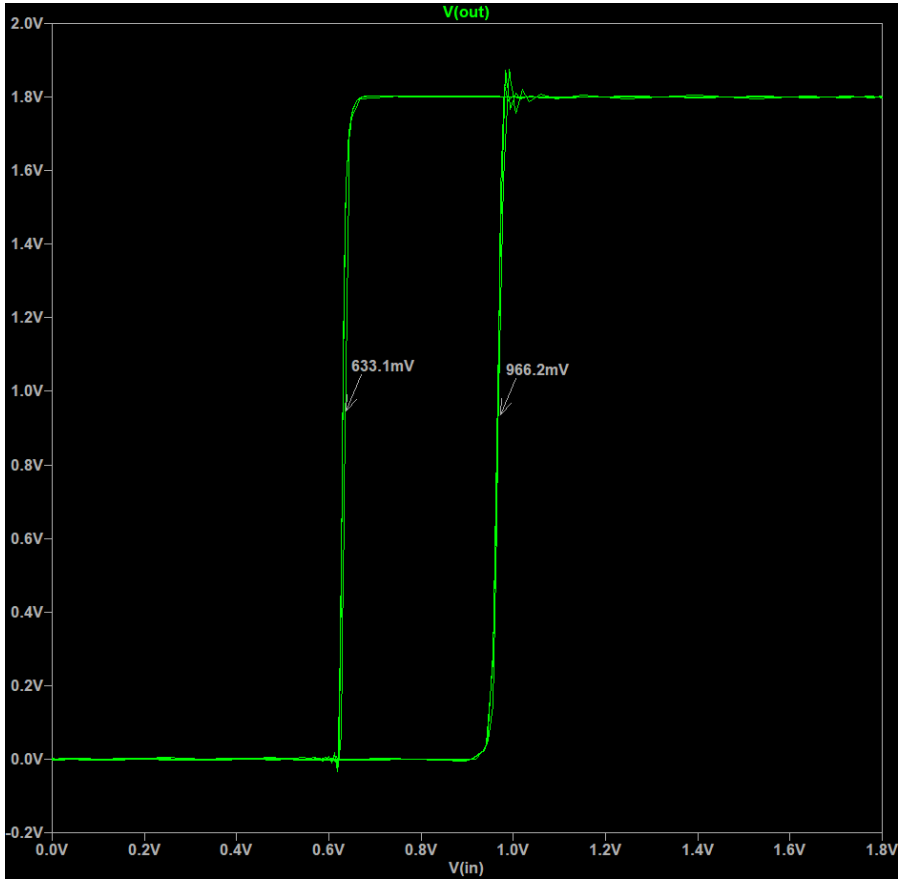
# Przerzutnik Schmitta (wersja nieodwracająca)

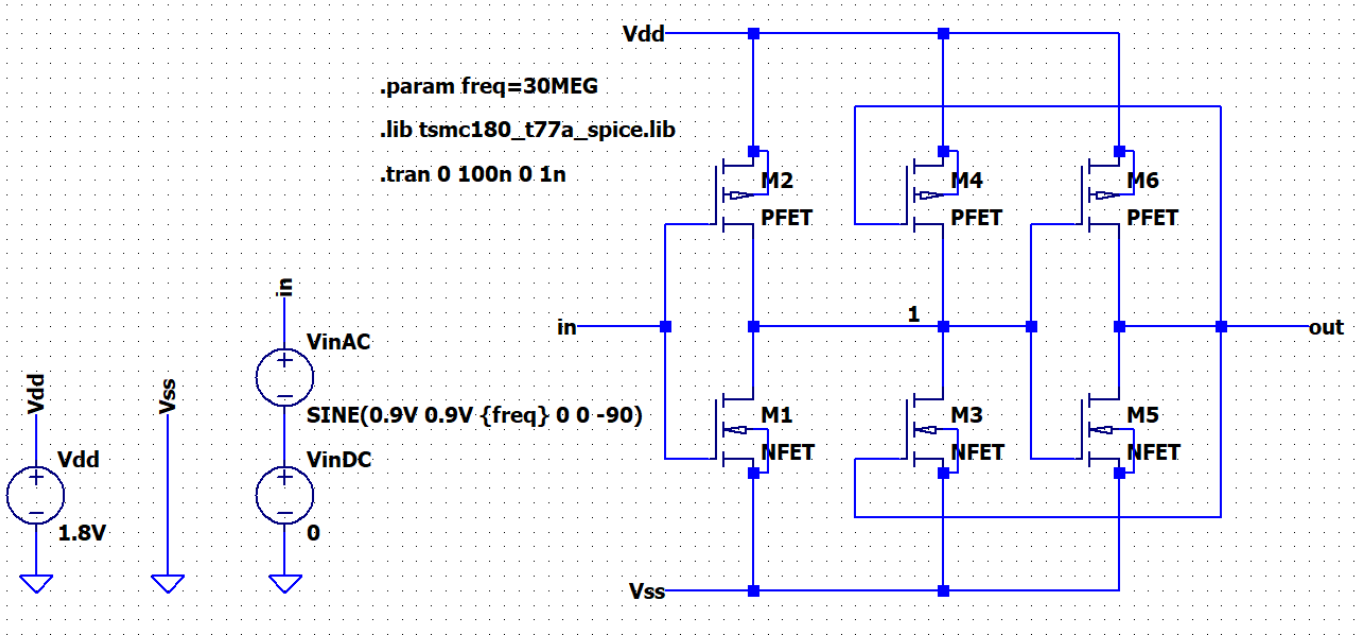



```
.subckt SCHMITT_TRIGGER in out vdd vss
M1 1 in vss vss NFET l=.2u w=2u
M2 vdd in 1 vdd PFET l=.2u w=4u
M3 1 out vss vss NFET l=.2u w=.4u
M4 vdd out 1 vdd PFET l=.2u w=.8u
M5 out 1 vss vss NFET l=.2u w=.4u
M6 vdd 1 out vdd PFET l=.2u w=.8u
.ends
```

# Pętla histerezy przerzutnika Schmitta

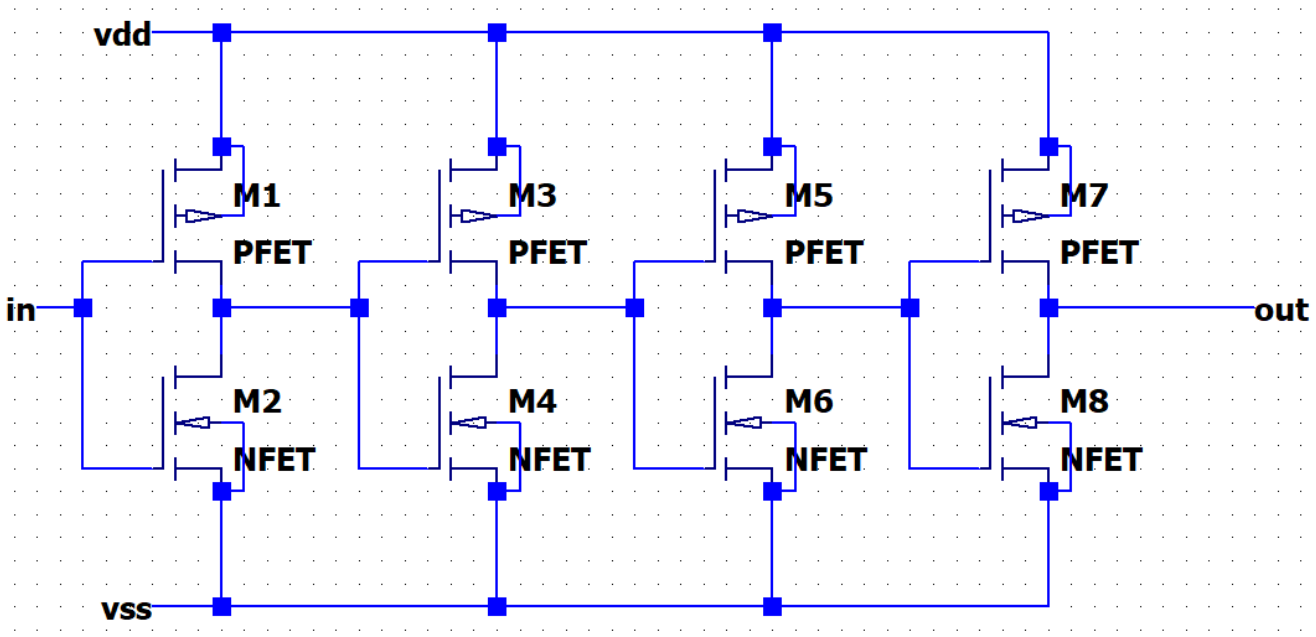



Vp = 966.2mV (napięcie przełączania z stanu niskiego na wysoki)
Vn =633.1mV (napięcie przełączania z stanu wysokiego na niski)
Vh = Vp – Vn = 333.1mV (szerokość pętli histerezy

# Bufor wyjściowy

```
.subckt OUTPUT_BUFFER in out vdd vss
M1 vdd in 1 vdd PFET l=.2u w=.8u
M2 1 in vss vss NFET l=.2u w=.4u
M3 vdd 1 2 vdd PFET l=.2u w=8.4u
M4 2 1 vss vss NFET l=.2u w=4.2u
M5 vdd 2 3 vdd PFET l=.2u w=89u
M6 3 2 vss vss NFET l=.2u w=44.5u
M7 vdd 3 out vdd PFET l=.2u w=938.7u
M8 out 3 vss vss NFET l=.2u w=469.4u
.ends
```



# Obliczenia teoretyczne skalowania bufora - klasyczna metoda skalowania

$$C_{ox} = \frac{\varepsilon_{ox}}{T_{ox}} = \frac{3{,}45 \cdot 10^{-11}\ F/m}{4{,}1 \cdot 10^{-9}\ m} \approx 8{,}415\ fF/\mu m^2$$

$$C_L = 25\,pF$$

$$C_G = C_{GN} + C_{GP} = 8{,}415\ fF/\mu m^2 \cdot 0{,}2\,\mu m \cdot (0{,}4\,\mu m + 0{,}8\,\mu m) =$$

$$= 2{,}02\ fF$$

przyjmujemy optymalną wartość $\alpha = e$

$$n = \frac{\ln\left(\frac{C_L}{C_G}\right)}{\ln \alpha} = \frac{\ln\left(\frac{25\,pF}{2{,}02\,fF}\right)}{\ln e} = 9{,}424$$

przyjmujemy $n = 9$

# Obliczenia teoretyczne skalowania bufora

$$\alpha = \left(\frac{C_L}{C_G}\right)^{\frac{1}{n}} = \left(\frac{25\,pF}{2{,}02\,fF}\right)^{\frac{1}{9}} = 2{,}849$$

ze względów praktycznych wybieramy mniejszą liczbę stopni $n = 4$

$$\alpha = \left(\frac{25\,pF}{2{,}02\,fF}\right)^{\frac{1}{4}} = 10{,}55$$

liczymy teoretyczną wartość maksymalnej częstotliwości bufora

$$K'_n = \mu_n \cdot C_{ox} = 289{,}44\ \frac{cm^2}{V\cdot s} \cdot 8{,}415\ \frac{fF}{\mu m^2} = 243\ \mu A/V^2$$

$$K'_p = \mu_p \cdot C_{ox} = 112{,}39\ \frac{cm^2}{V\cdot s} \cdot 8{,}415\ \frac{fF}{\mu m^2} = 94{,}5\ \mu A/V^2$$

# Obliczenia teoretyczne skalowania bufora

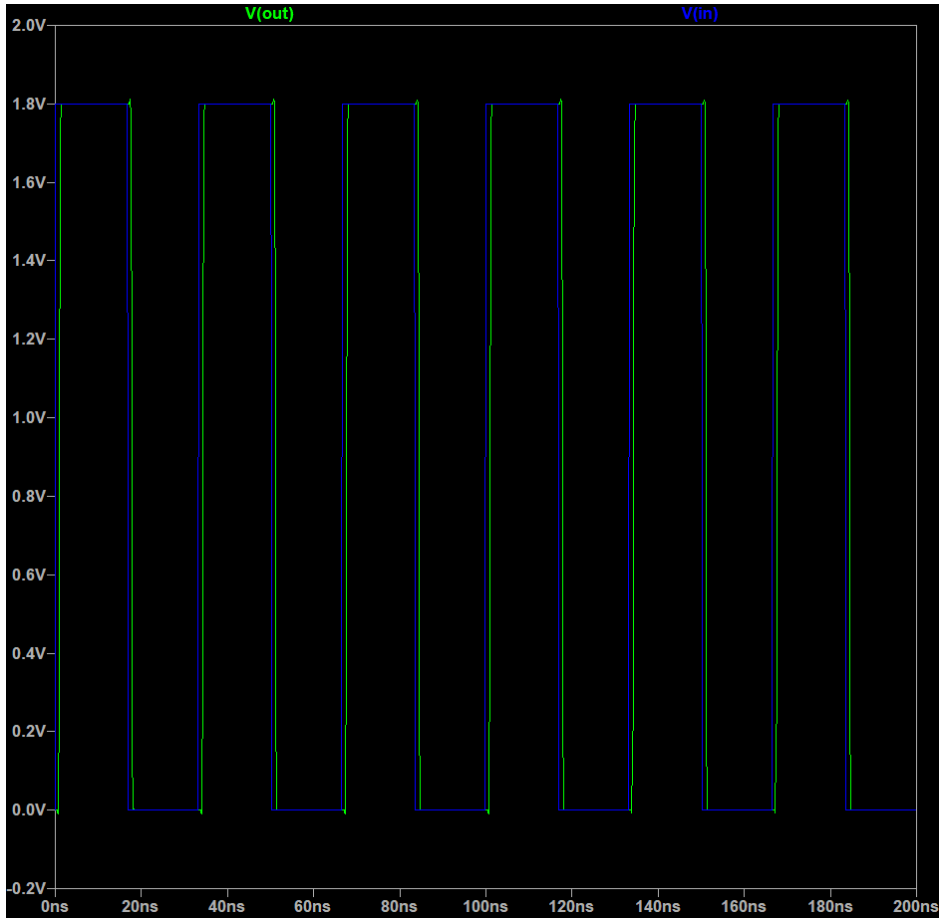
czas przejścia przez parę inwerterów:

$$t_{pid} = t_{LH} + t_{HL} = 2R_p C_G + 2R_N C_G = 26{,}77\,ps$$

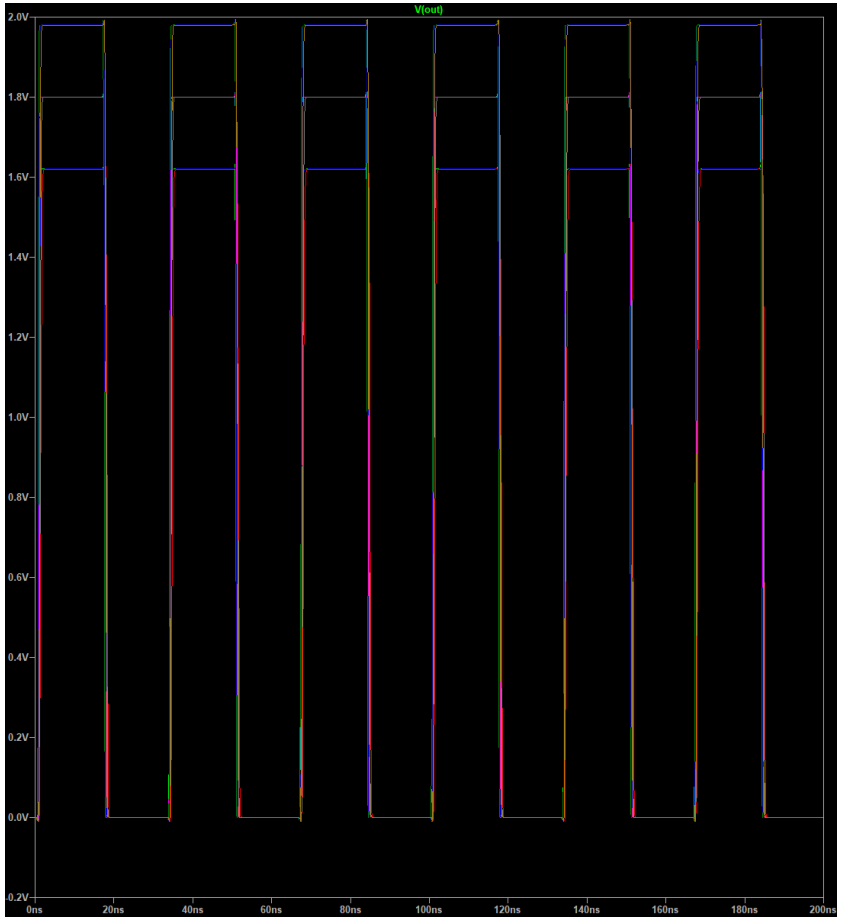
$$t_{cay} = n\alpha t_{apd} = n\alpha \frac{t_{pid}}{2} = 4 \cdot 10{,}55 \cdot \frac{26{,}77\,ps}{2} = 564{,}8\,ps$$

odpowiadająca częstotliwość $= \frac{1}{2 \cdot t_{cay}} = 885{,}3\,MHz$

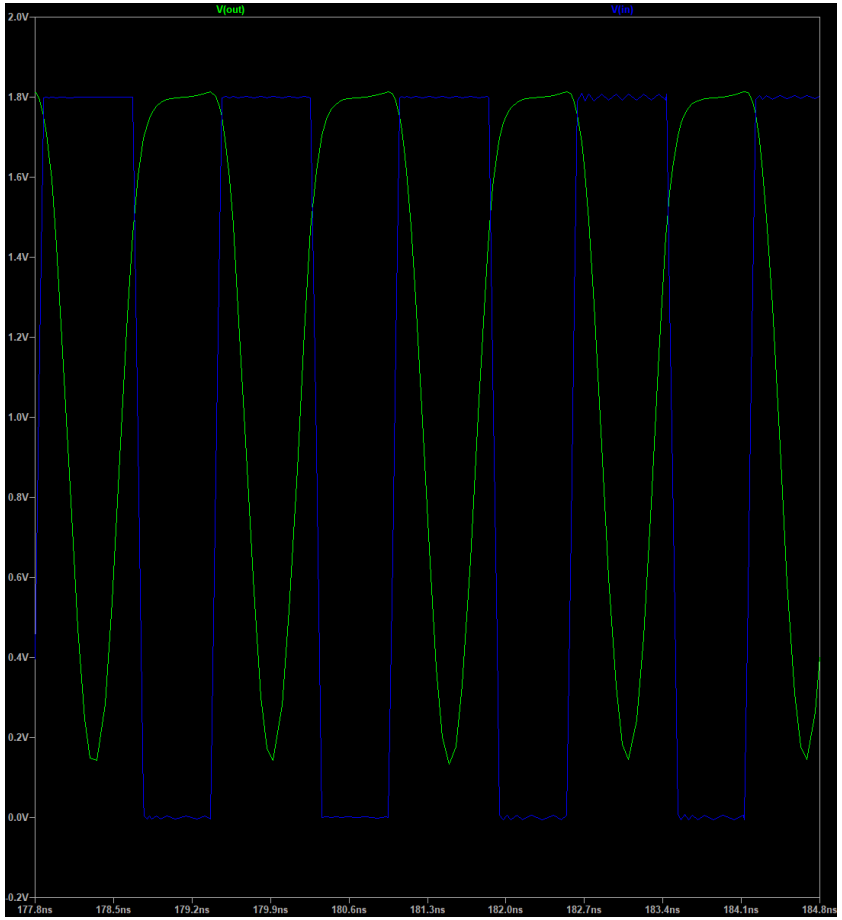
# Przebieg wyjściowy bufora przy częstotliwości 30MHz



# Przebieg wyjściowy bufora przy częstotliwości 30MHz dla różnych temperatur oraz napięć

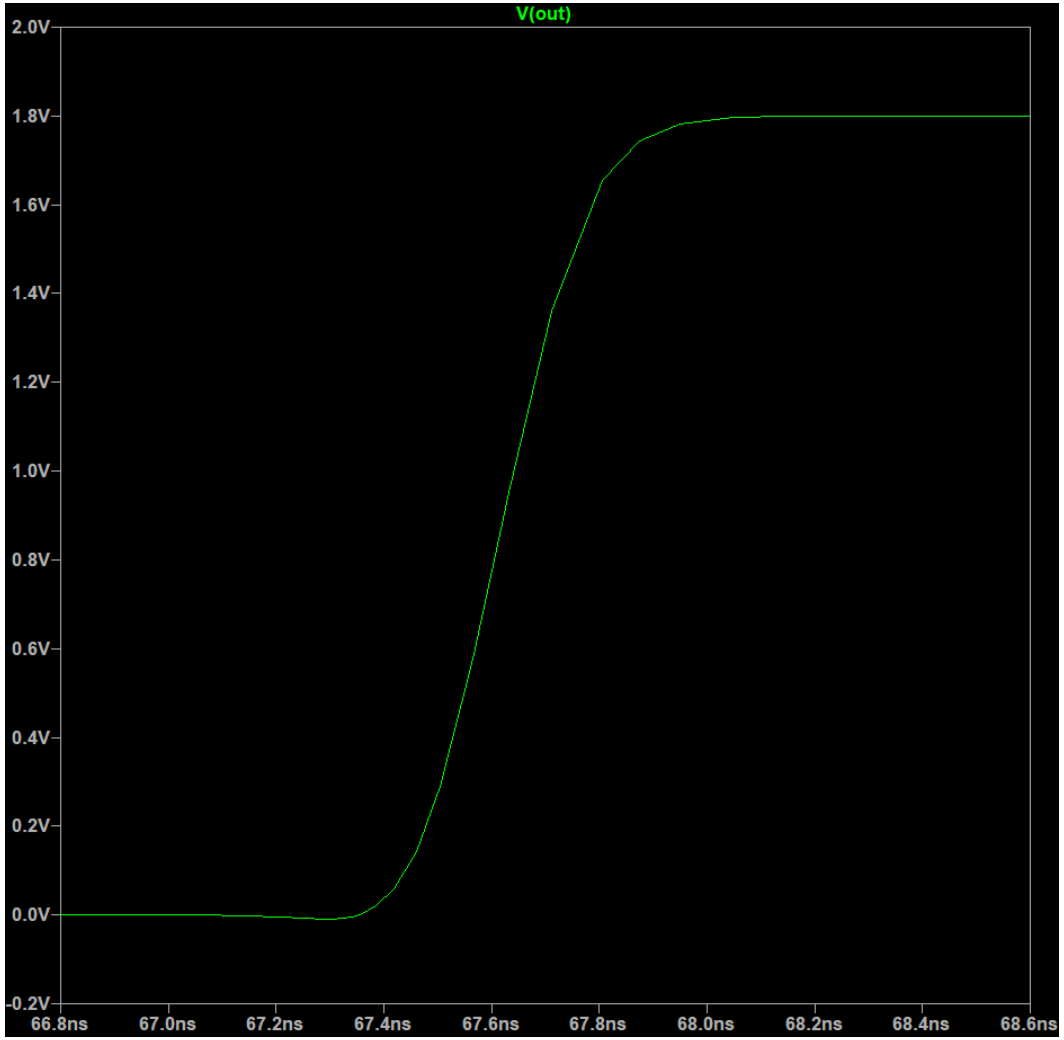


# Maksymalna częstotliwość  pracy bufora ok. 630MHz



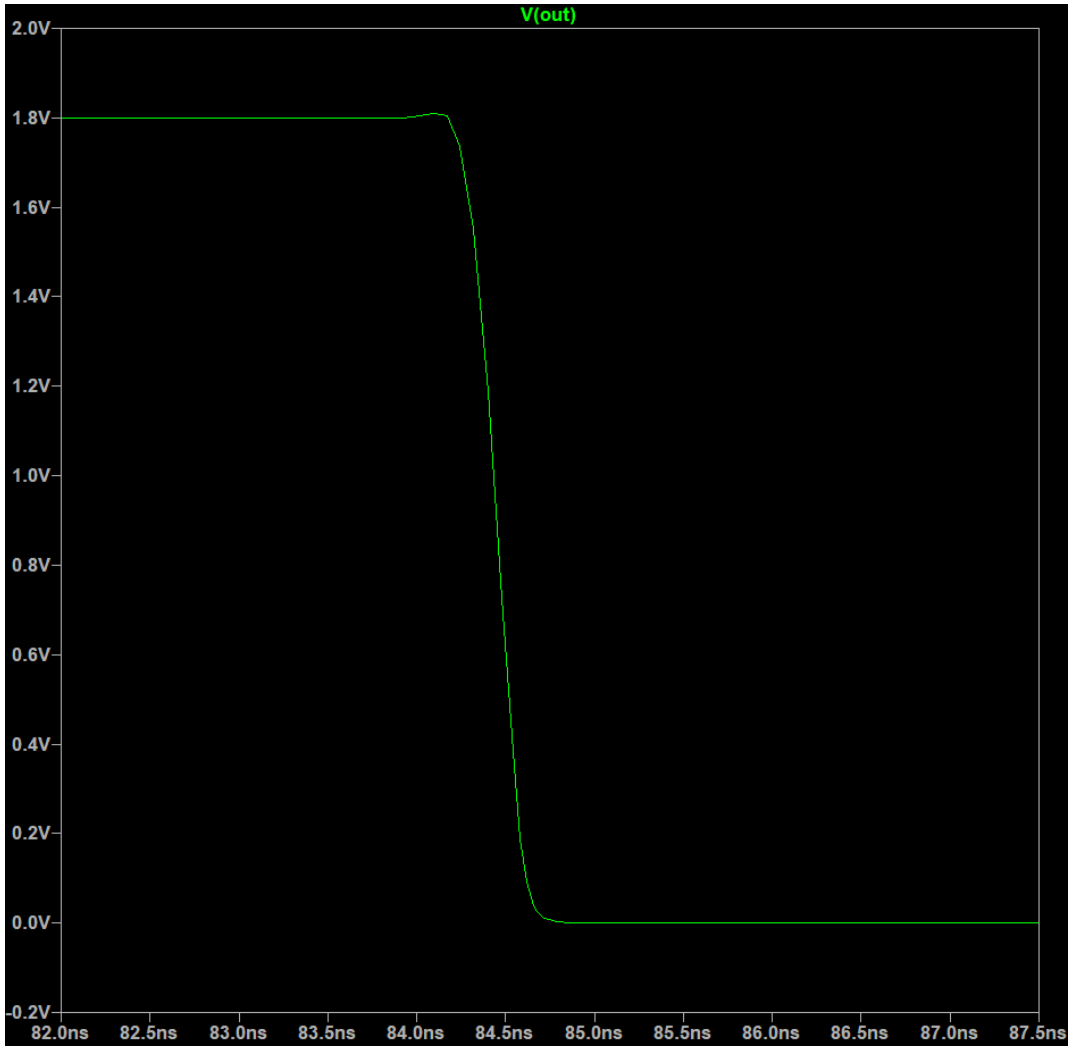
# Czas narastania

Czas narastania wynosi 326.1ps

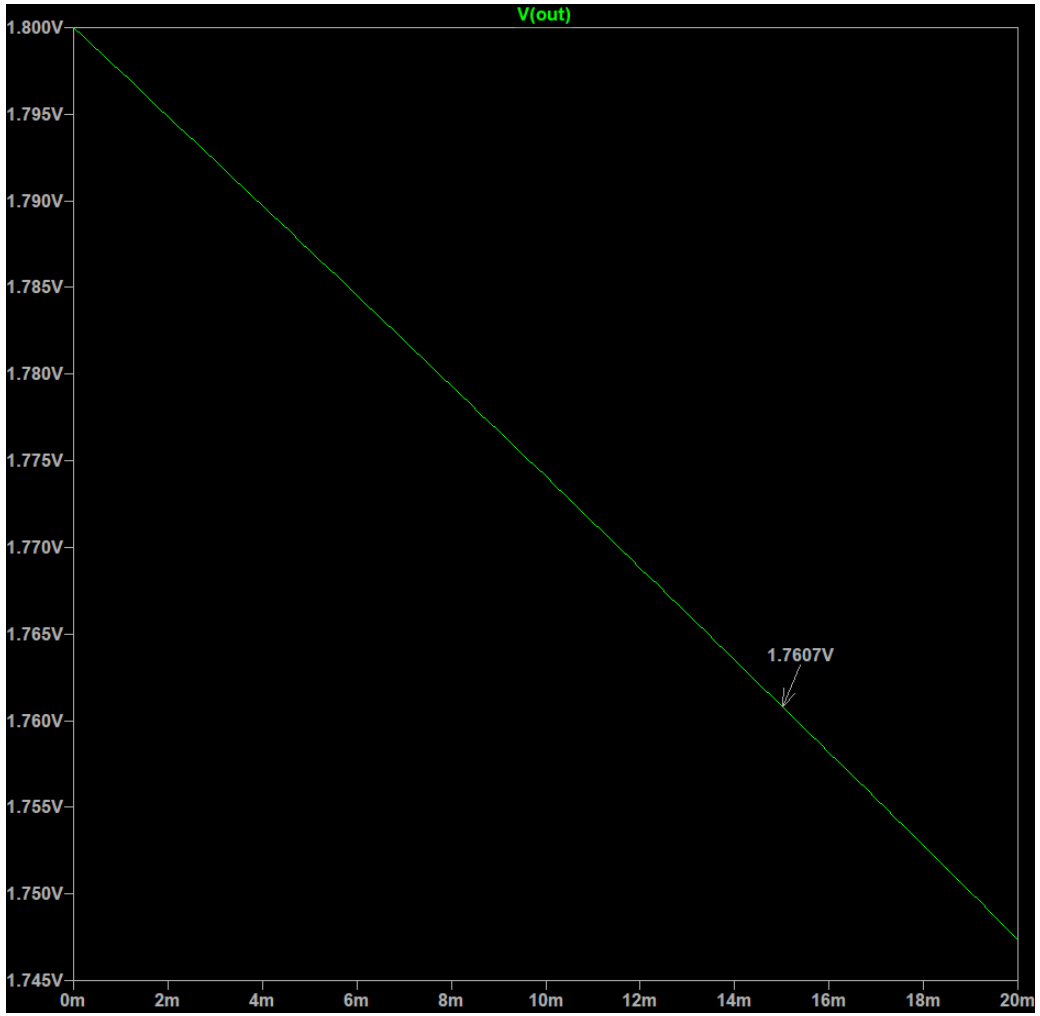


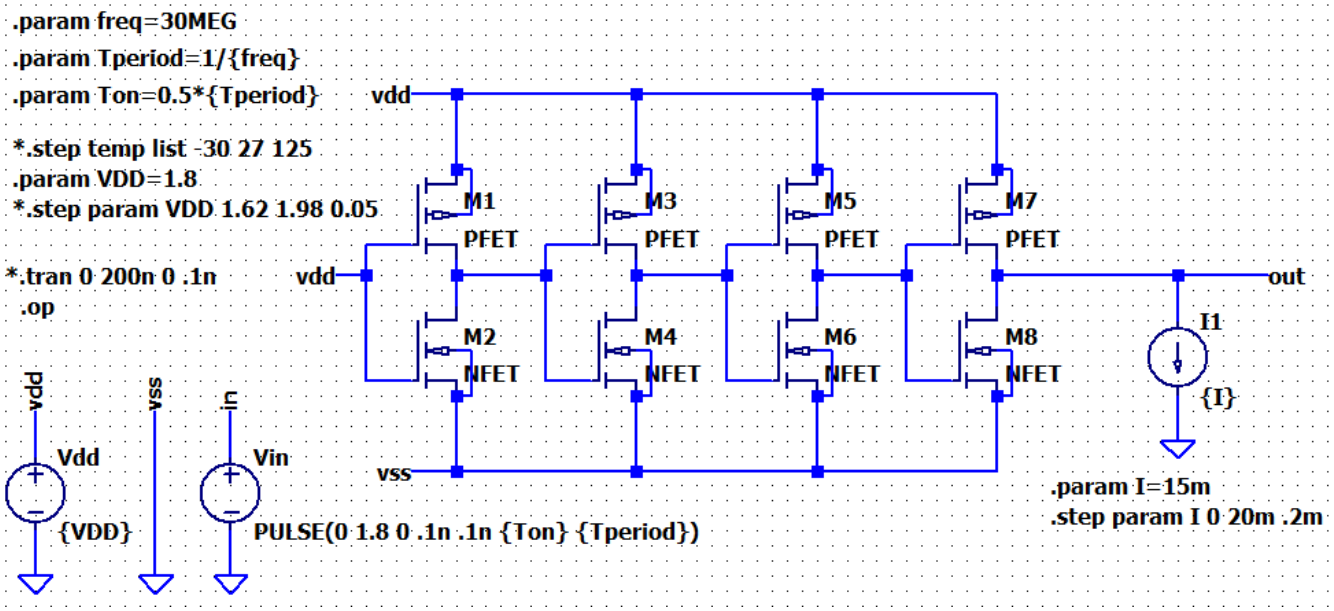
# Czas opadania

Czas opadania wynosi 313ps



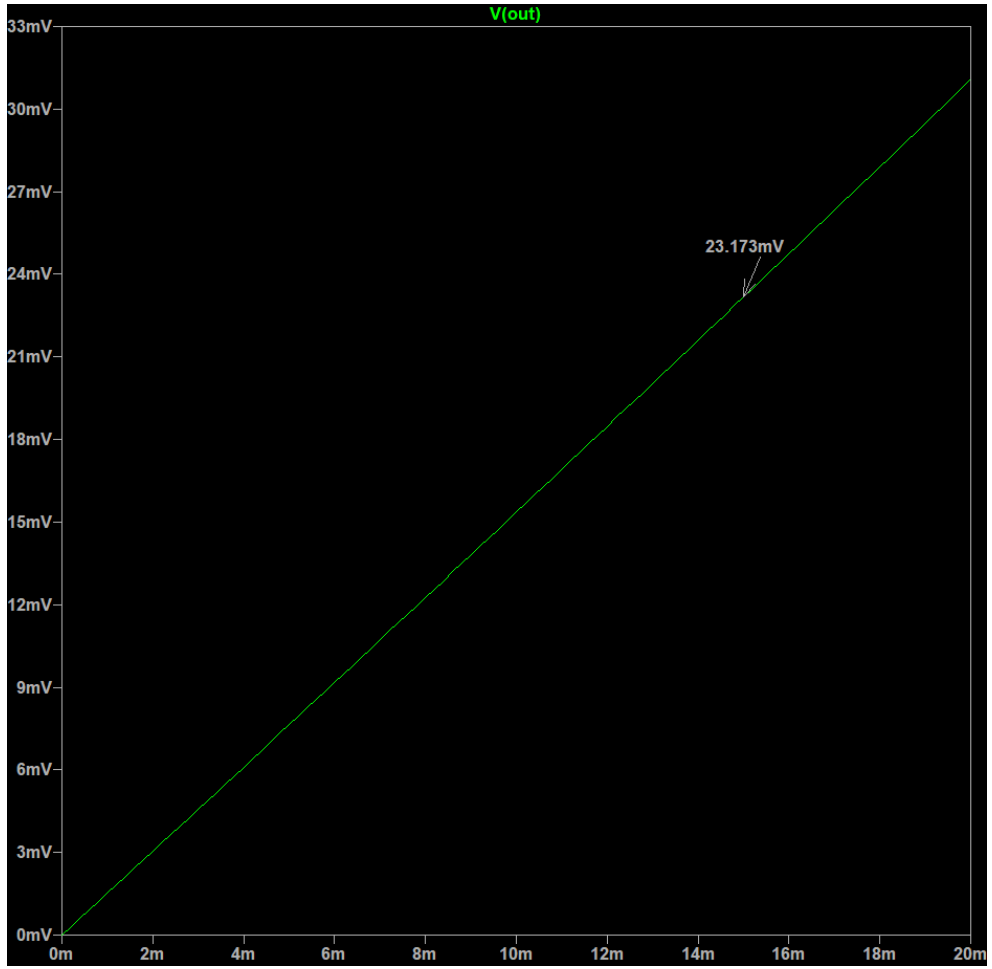
# Test wydajności prądowej bufora Io+

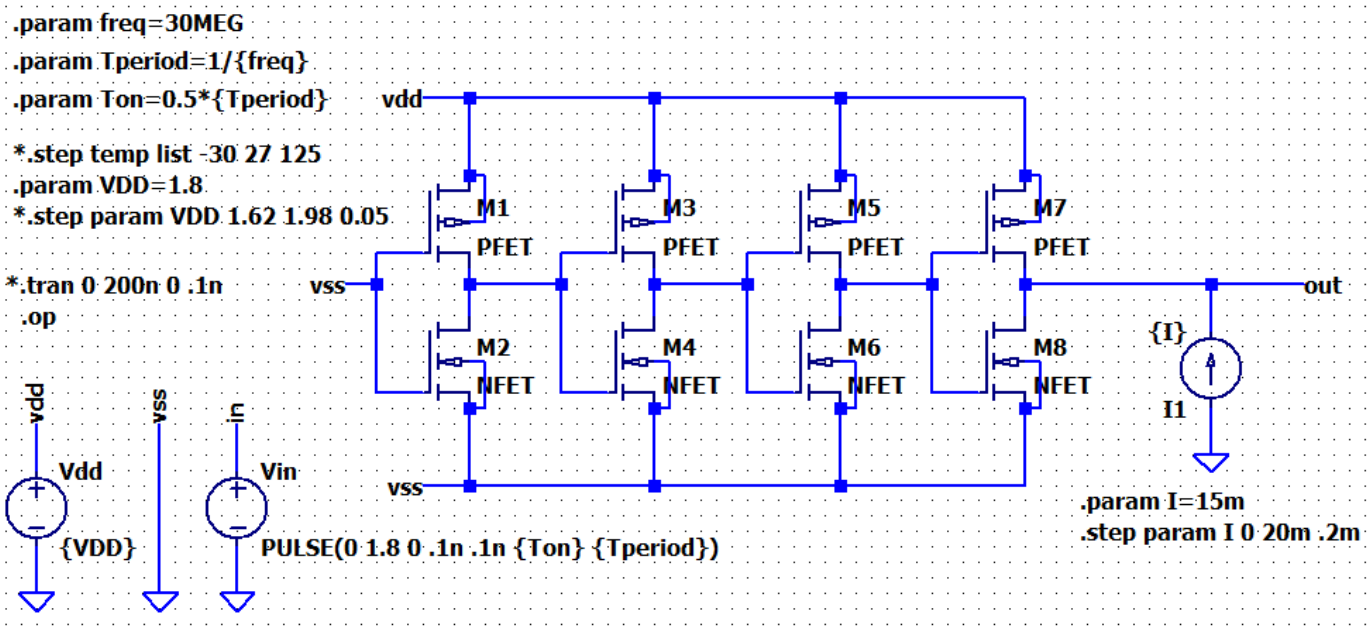



Vout dla Io+ = 15mA wynosi 1.76V

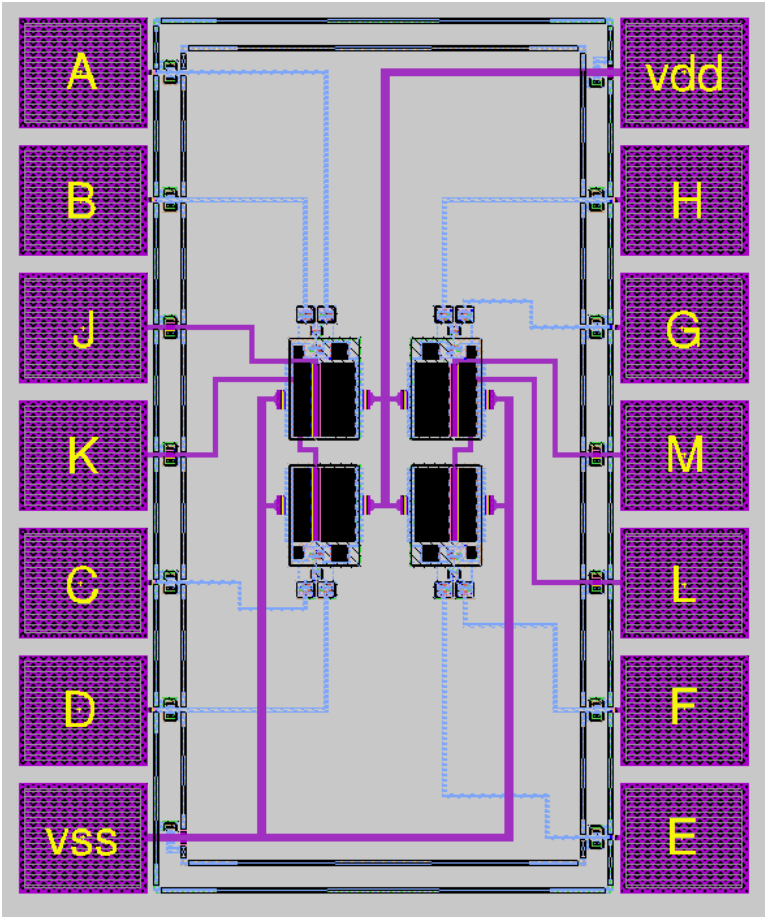
# Test wydajności prądowej bufora Io-

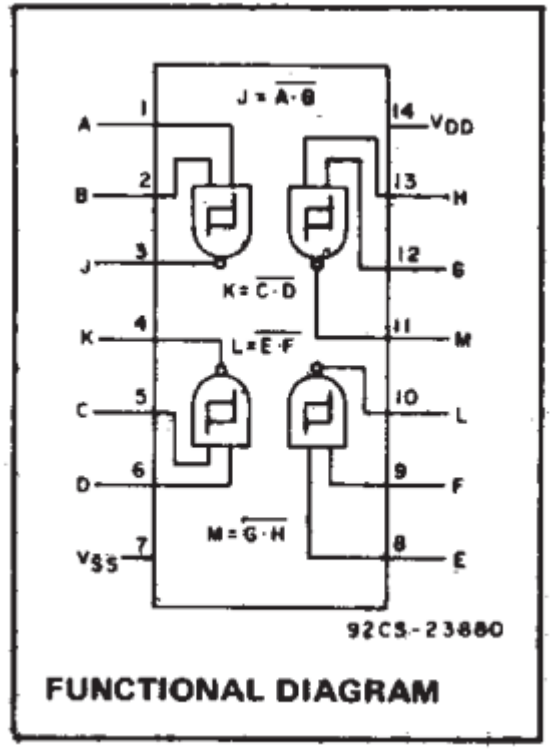


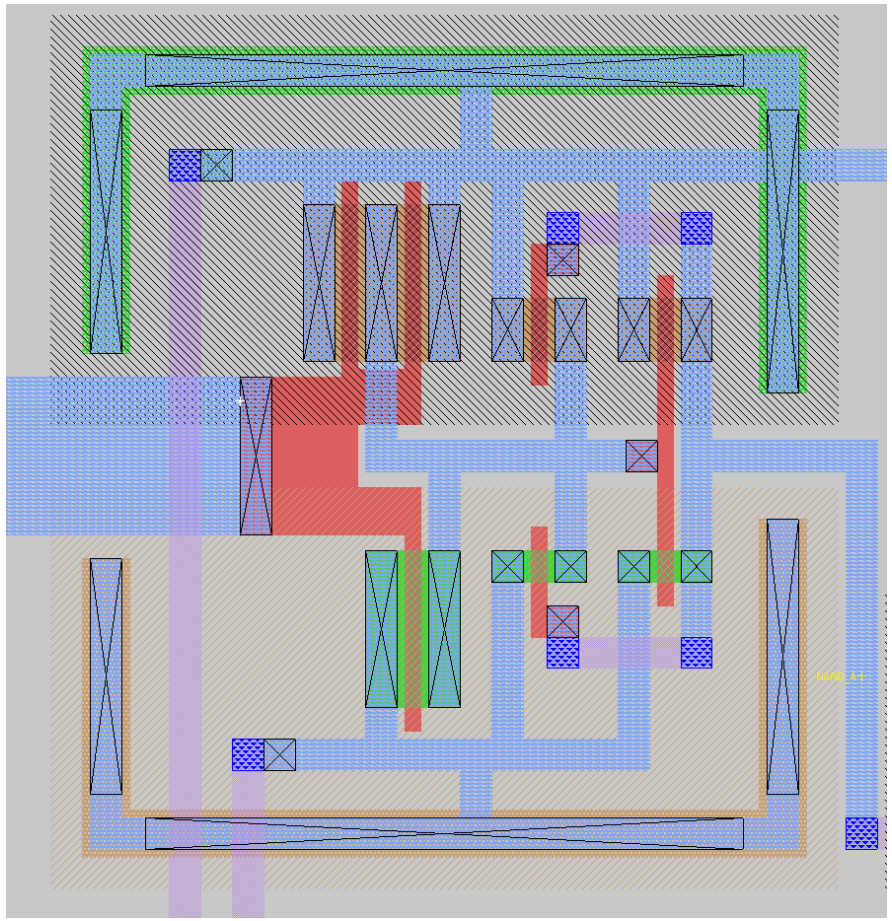

Vout dla Io- = 15mA wynosi 23.2mV

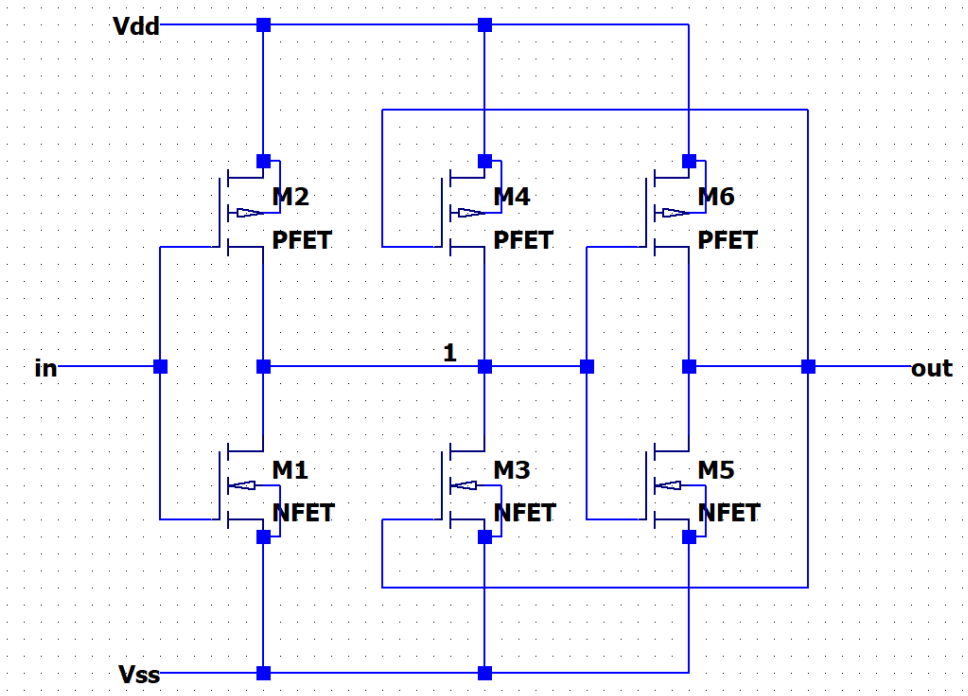
# Topografia całego układu



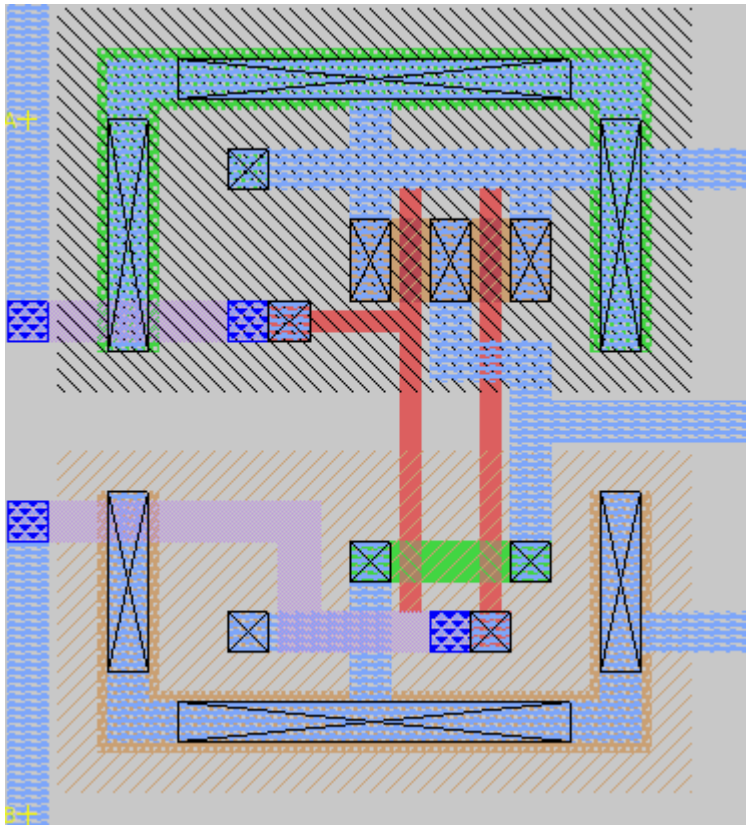



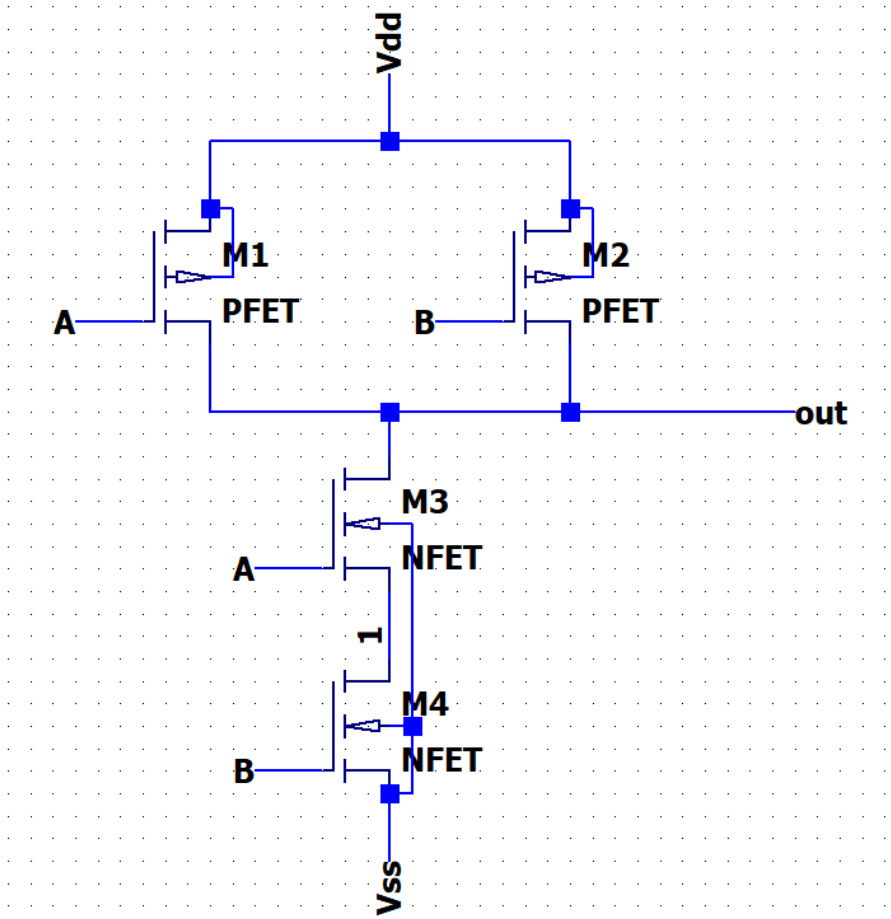
# Przerzutnik Schmitta



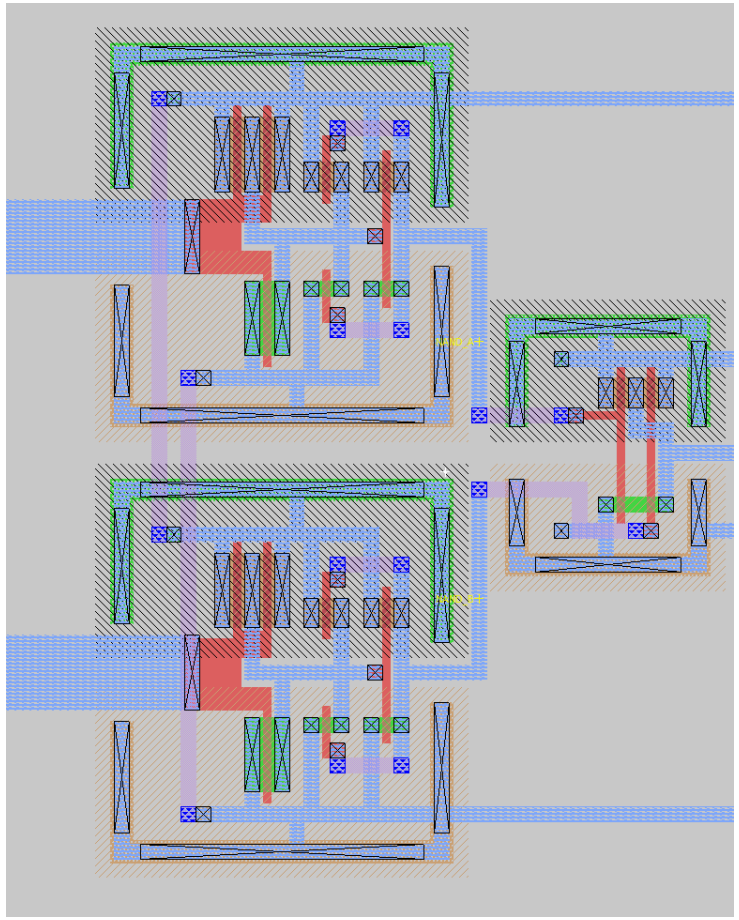


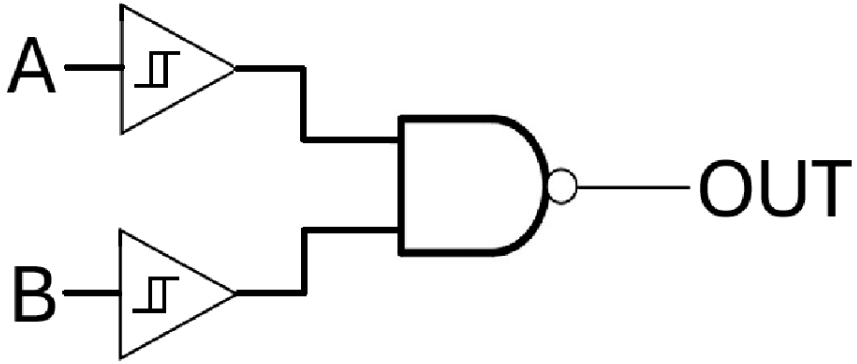
# Bramka NAND (bez histerezy)



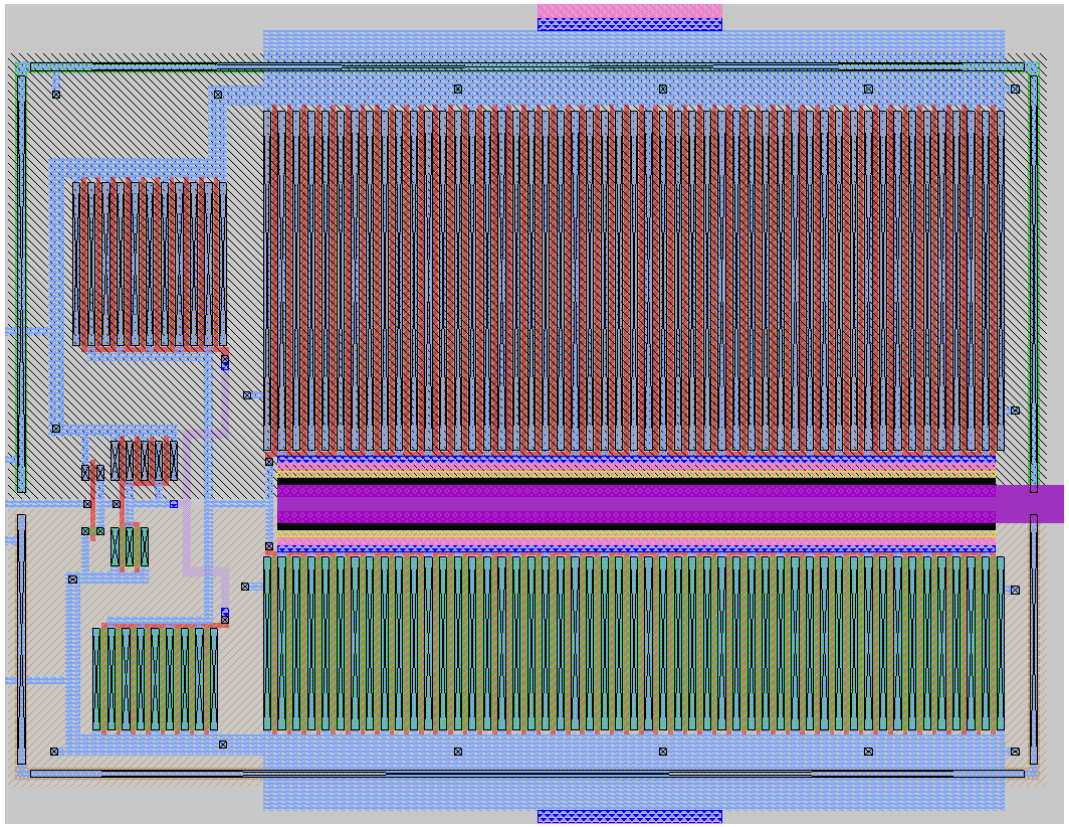


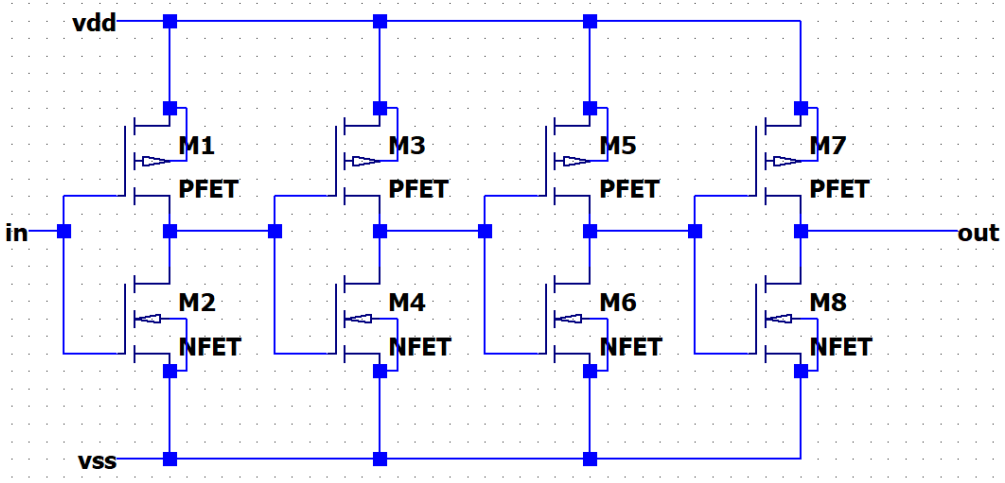
# Bramka razem z przerzutnikami



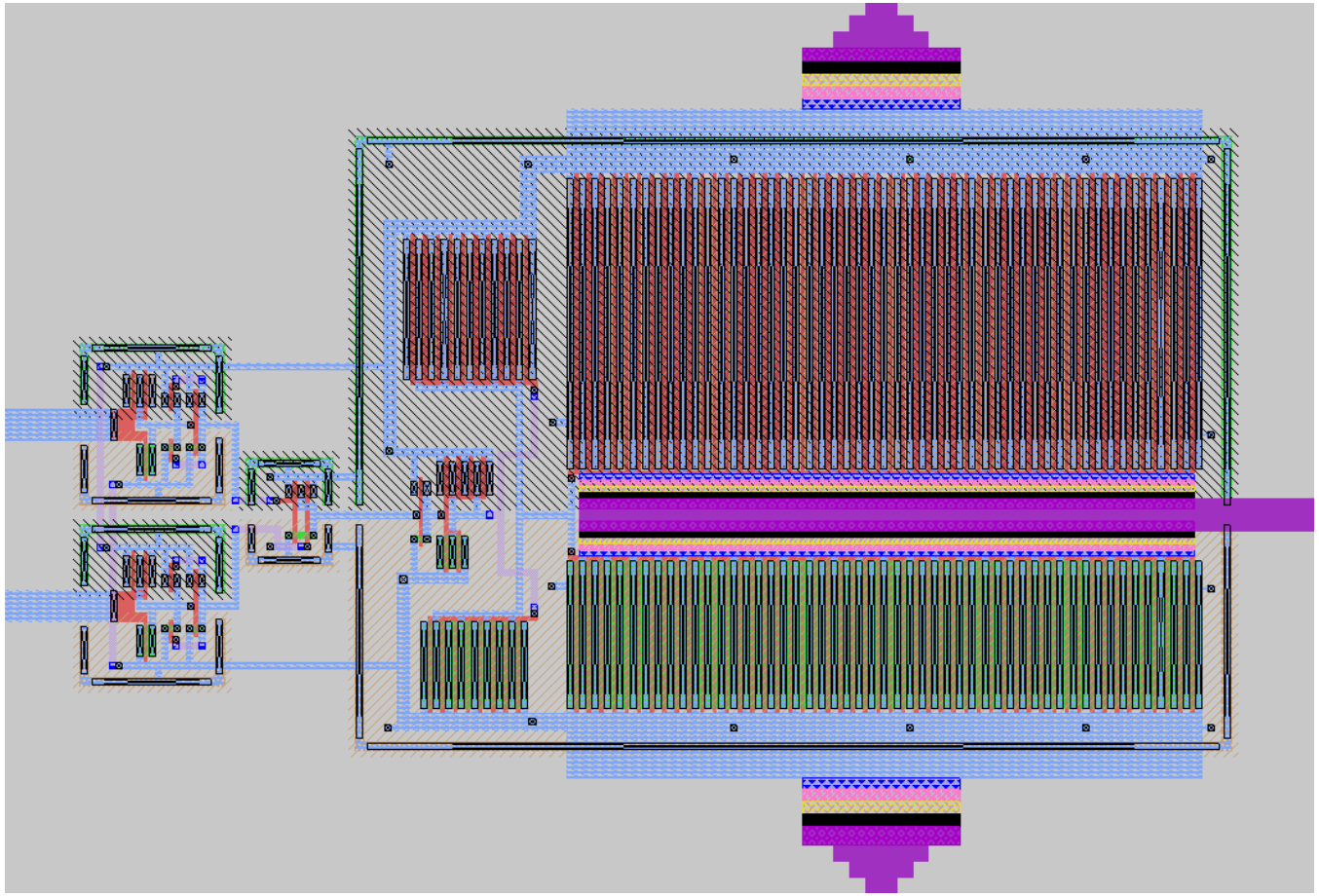


# Bufor wyjściowy

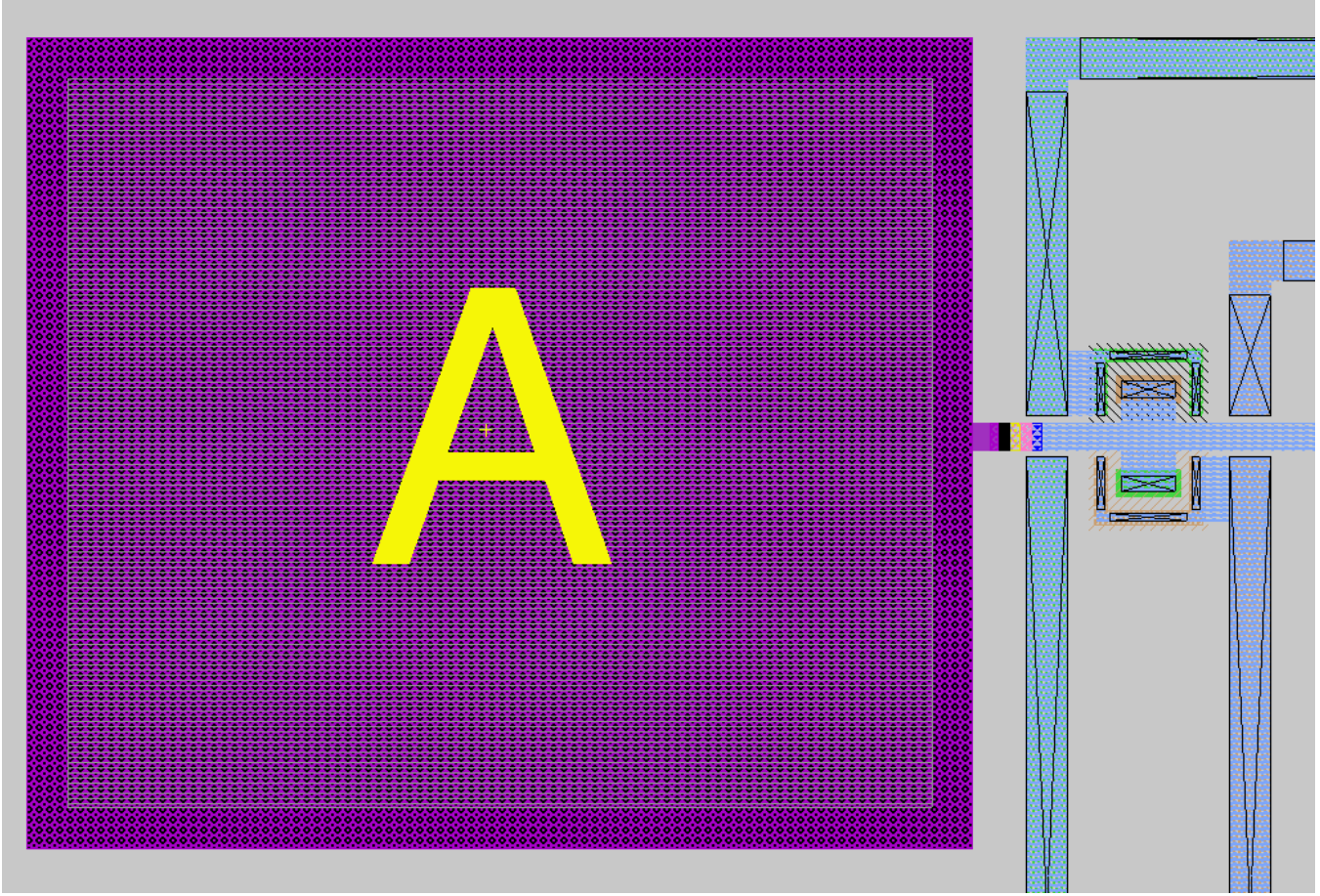




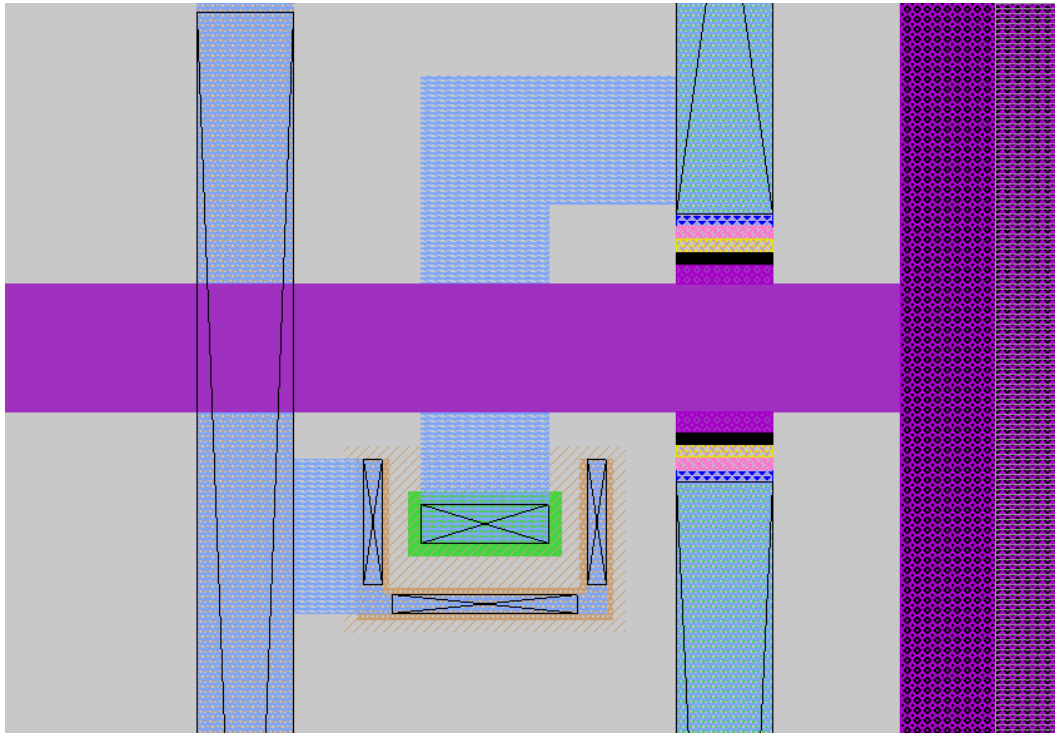
# Bramka wraz z buforem

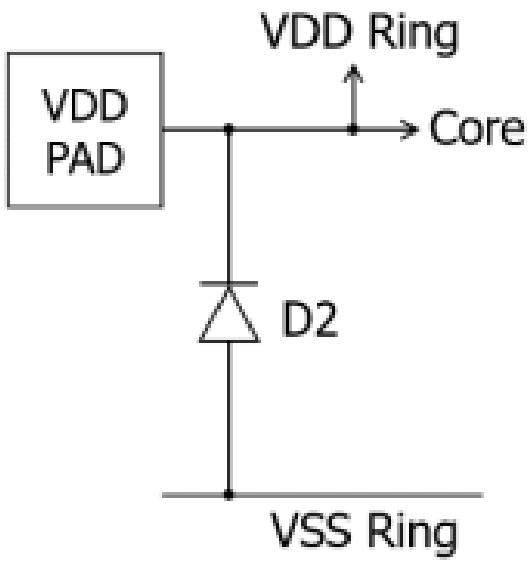


# Pole przyłączeniowe PAD

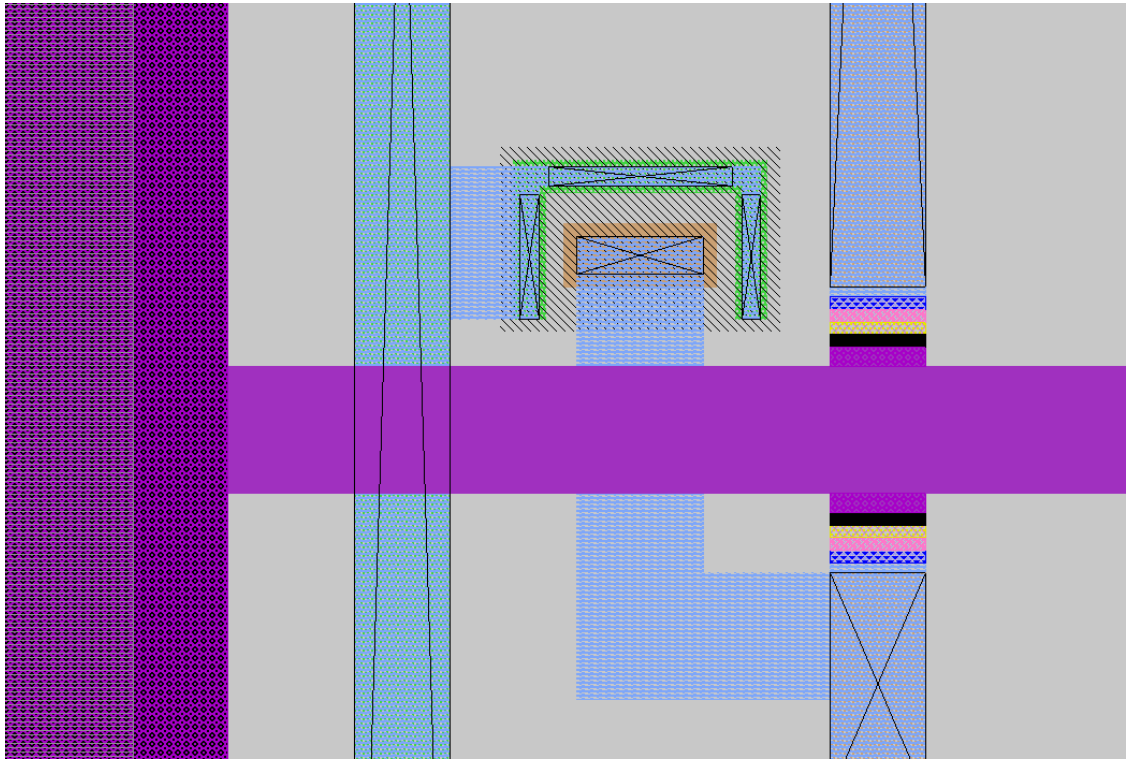


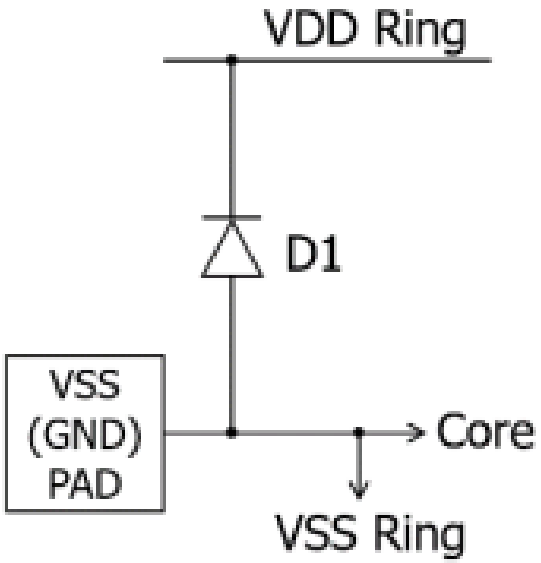
# Zabezpieczenie przed ESD (diodowe)



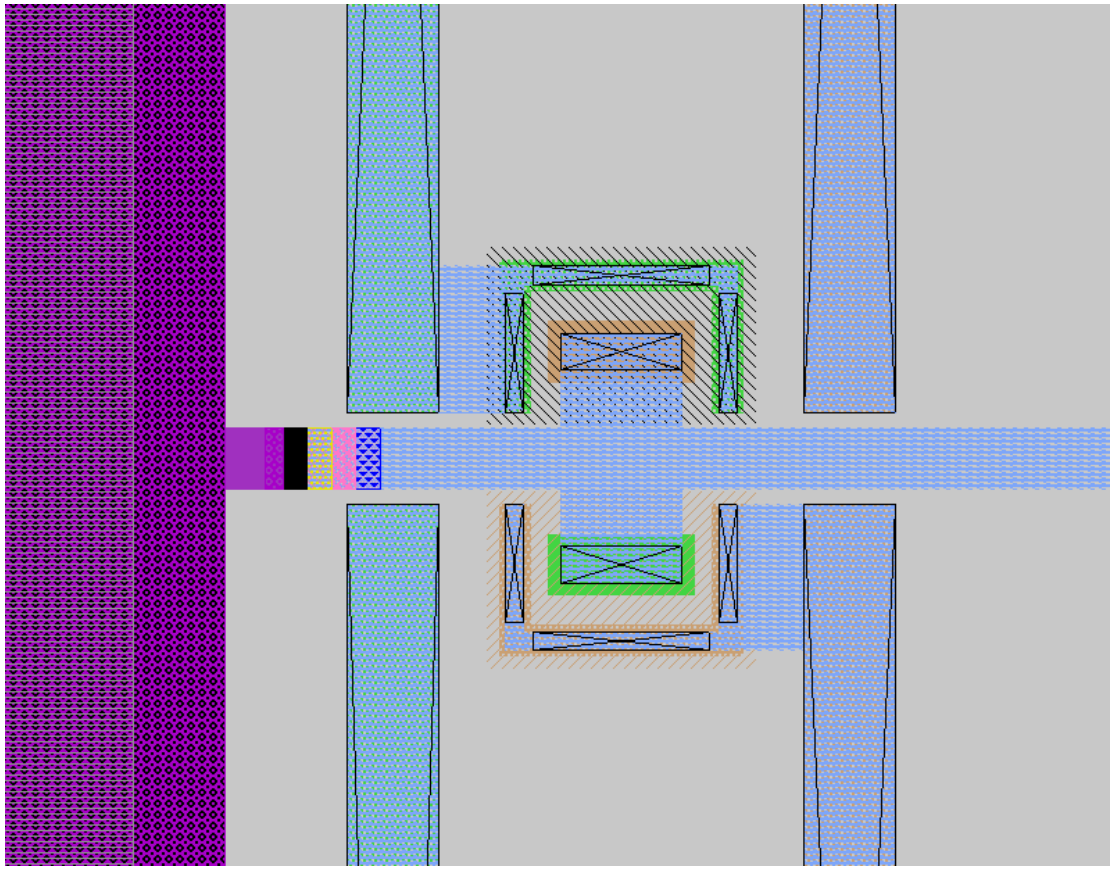


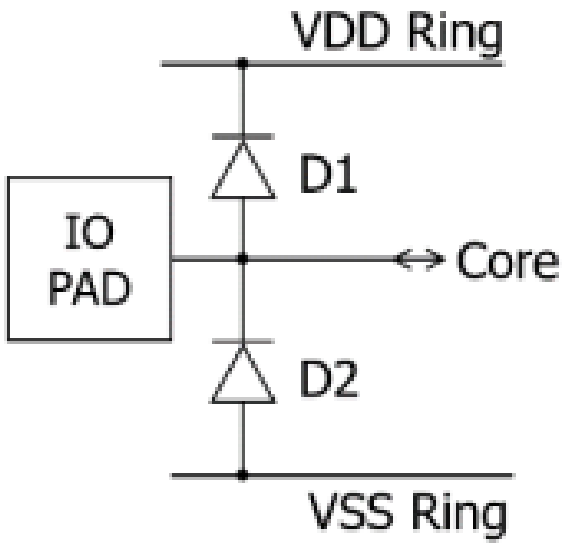
# Zabezpieczenie przed ESD (diodowe)



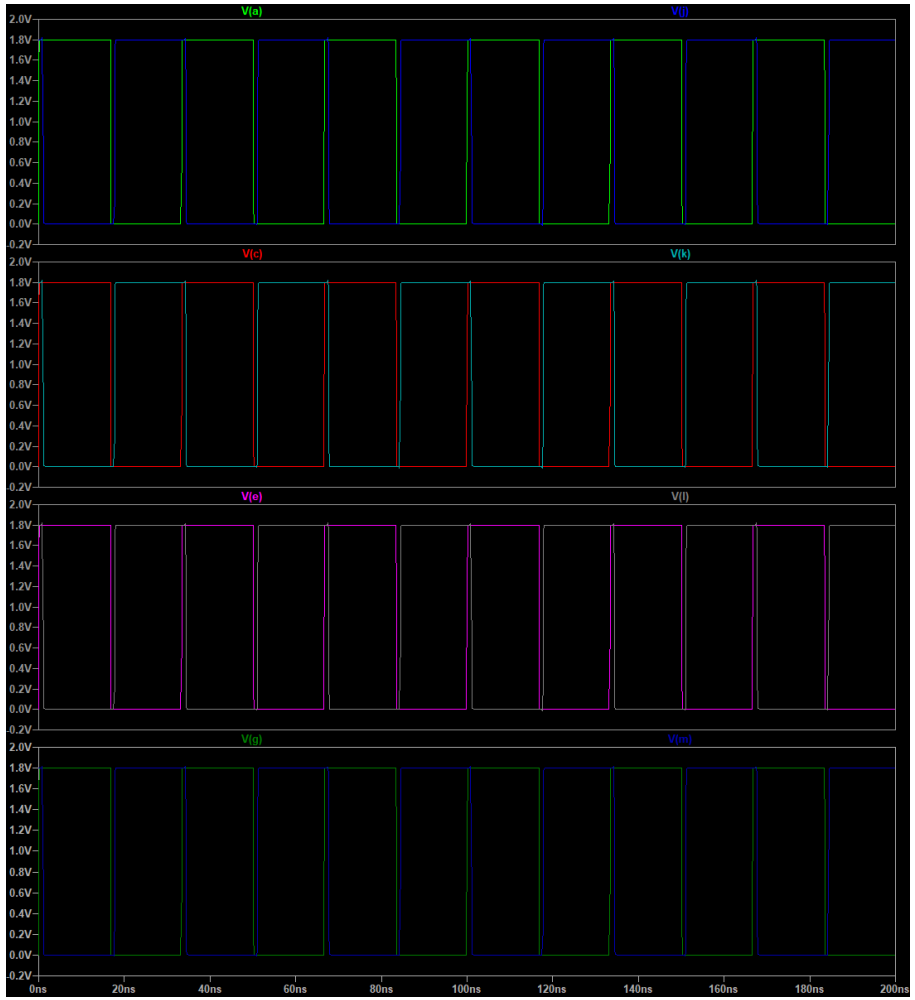


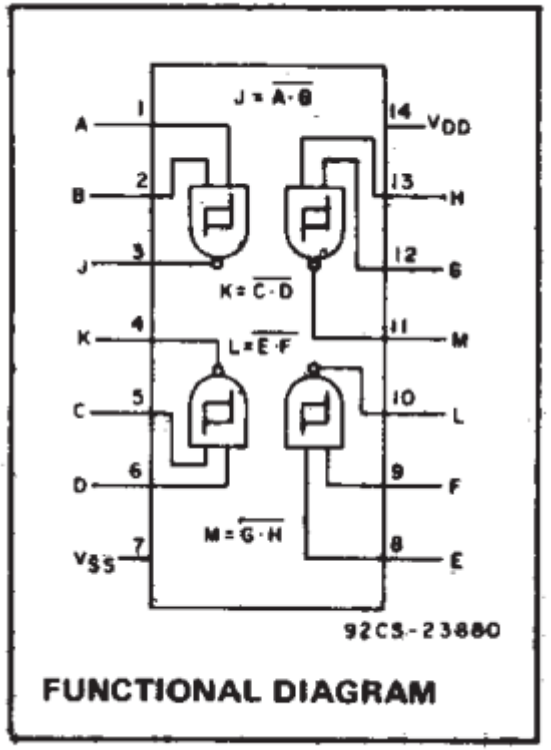
# Zabezpieczenie przed ESD (diodowe)





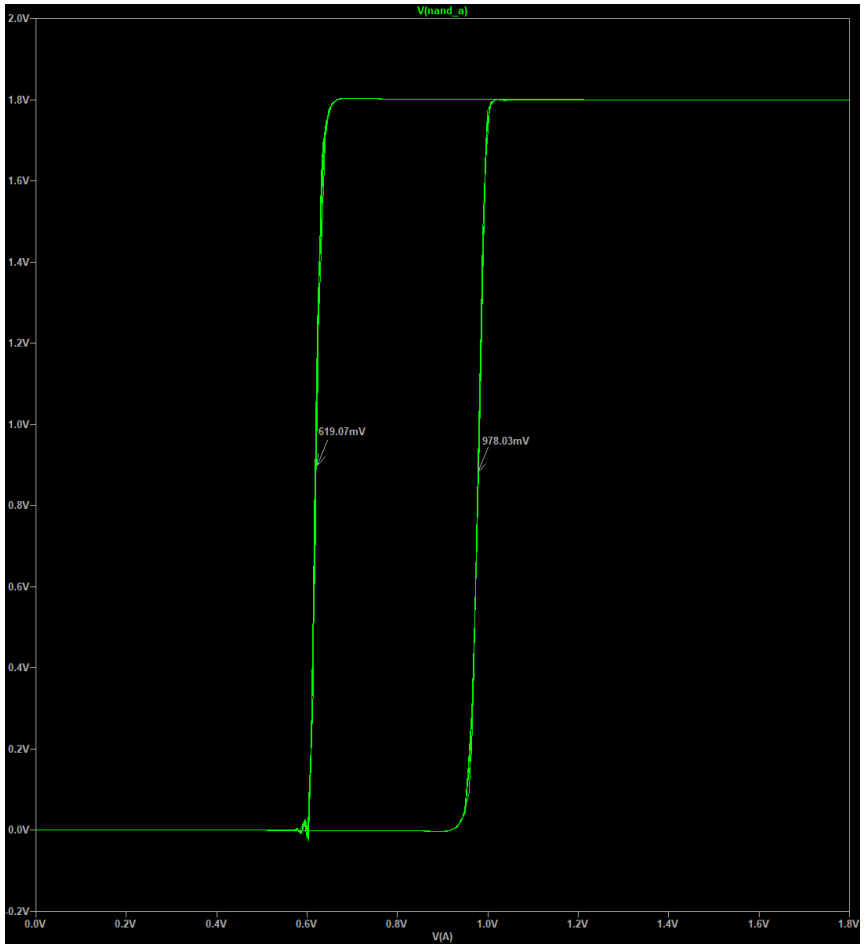
# Ogólny test działania układu po ekstrakcji



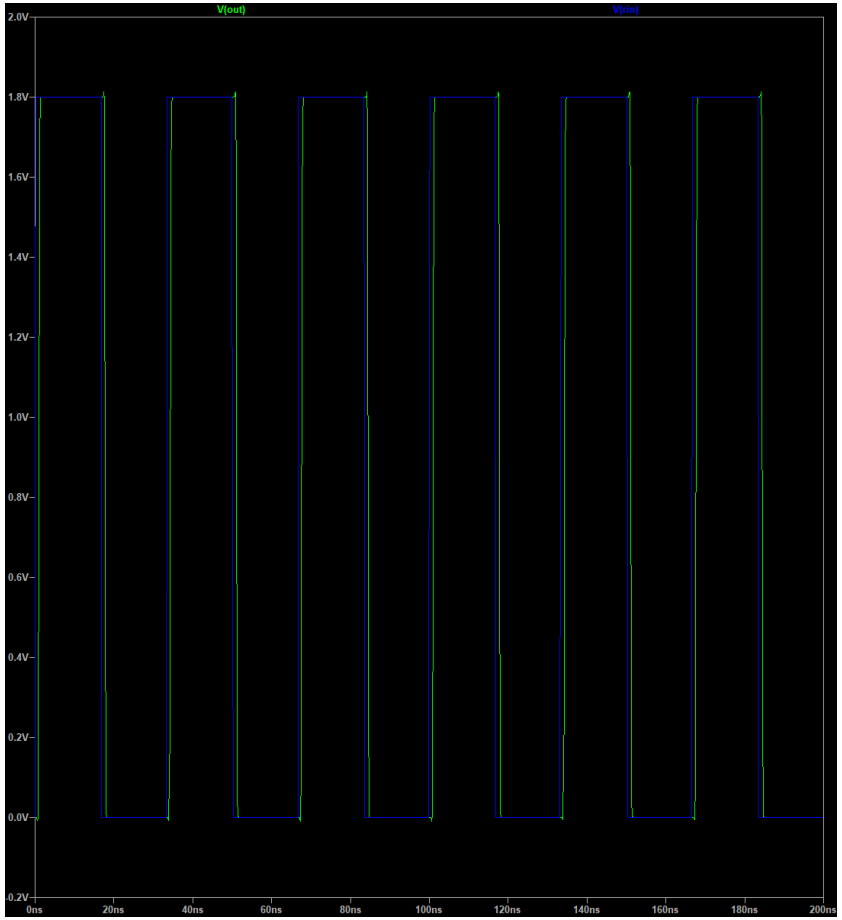

Piny A, C, E, G -> sygnał prostokątny 30MHz
Piny B, D, F, H -> Vdd

# Pętla histerezy przerzutnika Schmitta

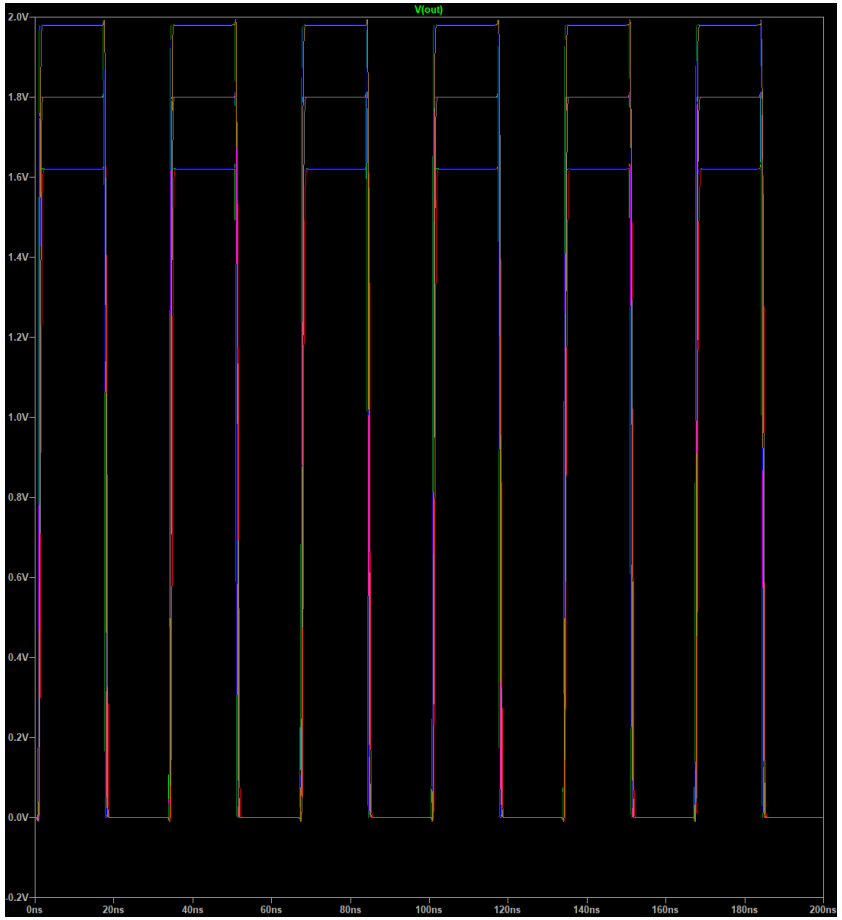



Pre-layout:
Vp = 966.2mV (napięcie przełączania z stanu niskiego na wysoki)
Vn =633.1mV (napięcie przełączania z stanu wysokiego na niski)
Vh = Vp – Vn = 333.1mV (szerokość pętli histerezy)
Post-layout:
Vp = 978.03mV
Vn = 619.07mV
Vh = 358.96mV

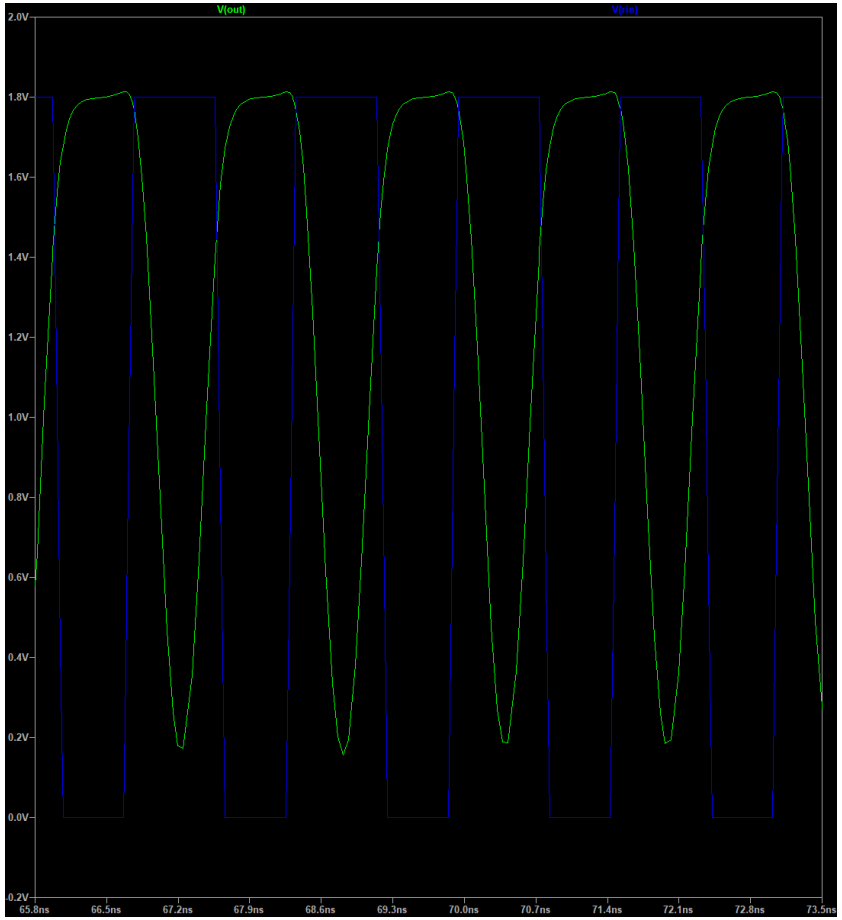
# Przebieg wyjściowy bufora przy częstotliwości 30MHz



# Przebieg wyjściowy bufora przy częstotliwości 30MHz dla różnych temperatur oraz napięć



# Maksymalna częstotliwość  pracy bufora ok. 630MHz (bez zmian)



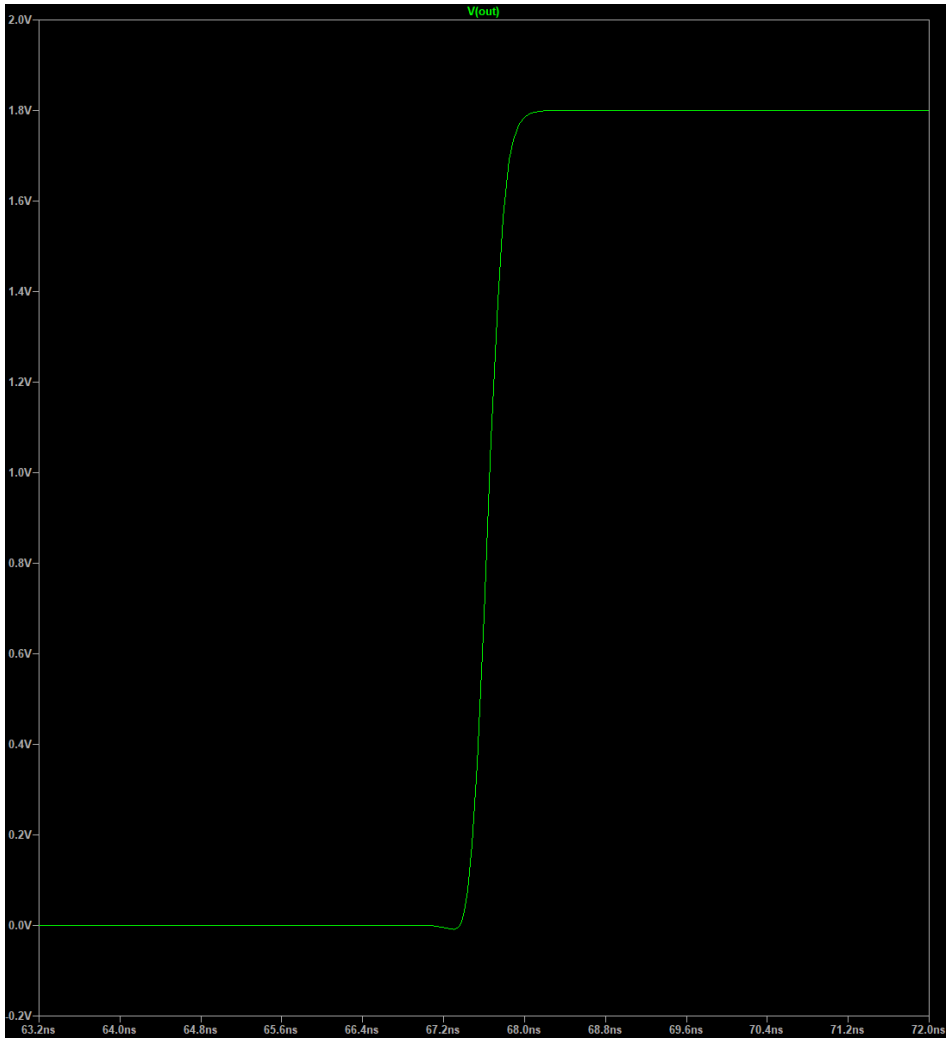
# Czas narastania

Pre-layout:

326.1ps

Post-layout:

332.5ps



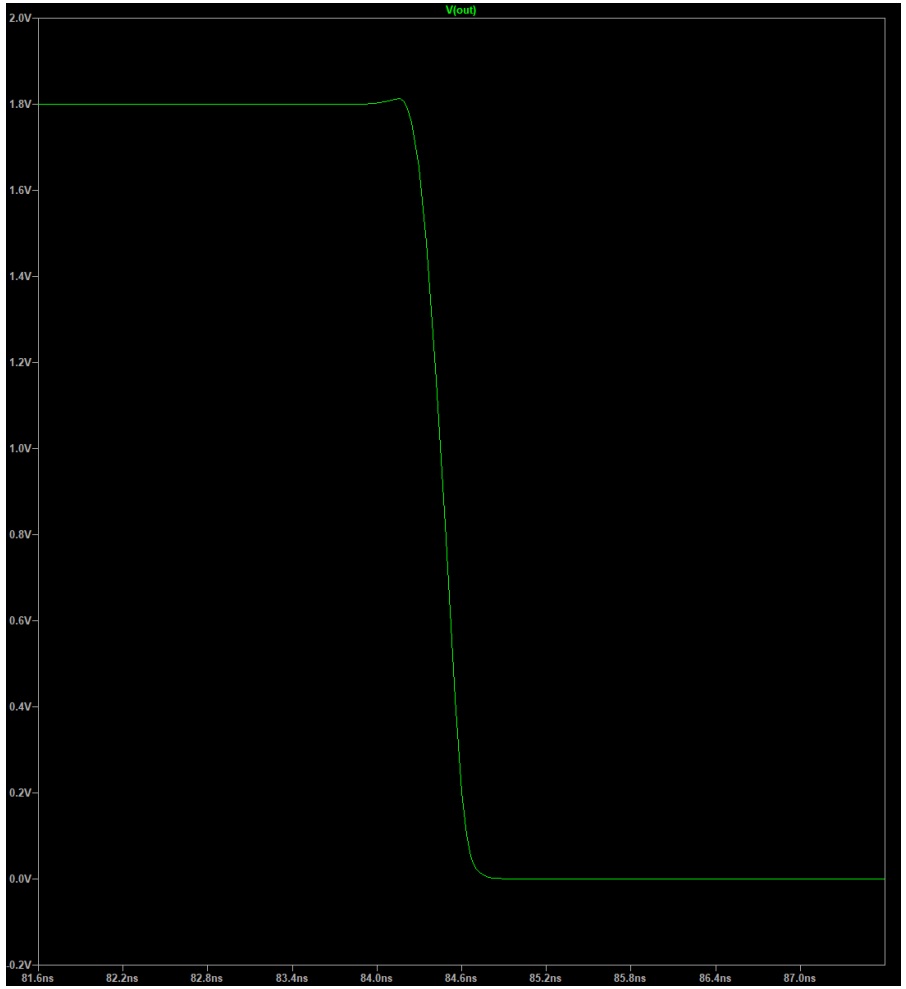
# Czas opadania

Pre-layout:

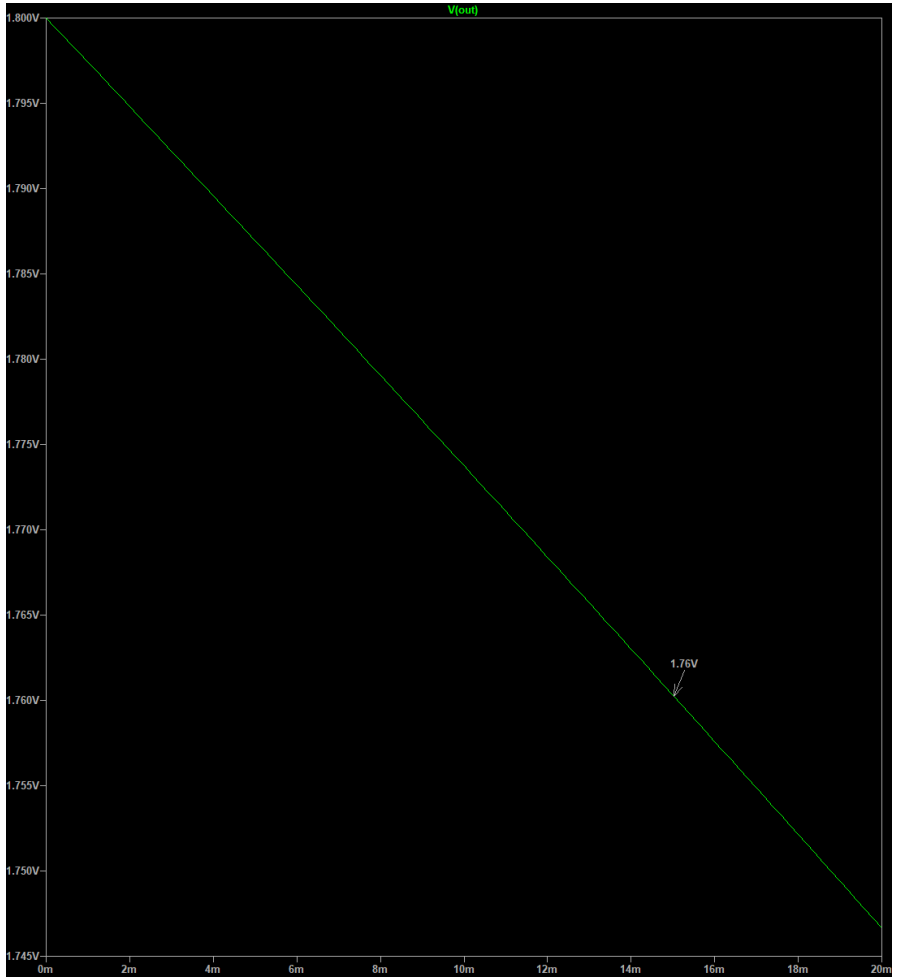
313ps

Post-layout:

302.3ps

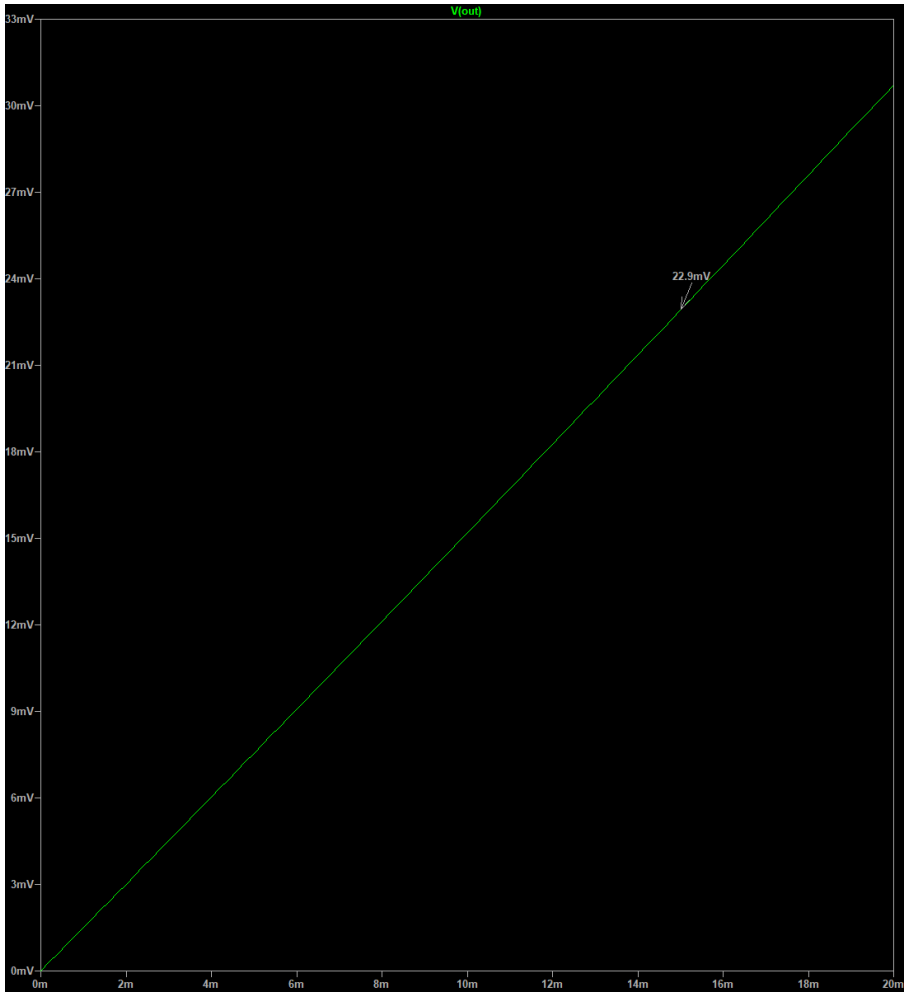


# Test wydajności prądowej bufora Io+




Pre-layout:
Vout dla Io- = 15mA wynosi 1.76V
Post-layout
Vout dla Io- = 15mA wynosi 1.76V

# Test wydajności prądowej bufora Io-




Pre-layout:
Vout dla Io- = 15mA wynosi 23.2mV
Post-layout
Vout dla Io- = 15mA wynosi 22.9mV